

Line Inputs

External Signal Levels

There are several standards for line signal levels. The -10 dBv standard is used for a lot of semi-professional recording equipment as it gives more headroom with unbalanced connections – the professional levels of $+4$ and $+6$ dBu assume balanced outputs that inherently give twice the output level for the same supply rails as it is measured between two pins with signals of opposite phase on them. See [Table 14.1](#)

Signal levels in dBu are expressed with reference to 0 dBu = 775 mVrms; the origin of this odd value is that it gives a power of 1 mW in a purely historical $600\ \Omega$ load. The unit of dBm refers to the same level but takes the power rather than the voltage as the reference – a distinction of little interest nowadays. Signals in dBv (or dBV) are expressed with reference to 0 dB = 1.000 Vrms.

These standards are well established, but that does not mean all equipment follows them. To take a current example, the Yamaha P7000S power amplifier requires $+8$ dBu (1.95 Vrms) to give its full output of 750 W into $8\ \Omega$.

Internal Signal Levels

In any audio system it is necessary to select a suitable nominal level for the signal passing through it. This level is always a compromise – the signal level should be high so it suffers minimal degradation by the addition of circuit noise as it passes through the system, but not so high that it is likely to suffer clipping before the gain control, or generate undue distortion

TABLE 14.1 Nominal signal levels

	Vrms	dBu	dBv
Semi-professional	0.316	-7.78	-10
Professional	1.228	$+4.0$	$+1.78$
German ARD	1.55	$+6.0$	$+3.78$

below the clipping level. (This last constraint is not normally a problem with modern circuitry, which gives very low distortion right up to the clipping point.)

It must always be considered that the gain control may be maladjusted by setting it too low and turning up the input level from the source equipment, making input clipping more likely. The internal levels chosen are usually in the range -6 to 0 dBu (388 mVrms to 775 Vrms) but in some specialized equipment such as broadcast mixing consoles, where levels are unpredictable and clipping distortion less acceptable than a bit more noise, the nominal internal level may be as low as -16 dBu (123 mVrms). If the internal level is in the normal -6 to 0 dBu range, and the maximum output of an op-amp is taken as 9 Vrms ($+21.3$ dBu), this gives 20–27 dB of headroom before clipping occurs.

If the incoming signal does have to be amplified, this should be done as early as possible in the signal path, to get the signal well above the noise floor as quickly as possible. If the gain is implemented in the first stage (i.e. the input amplifier, balanced or otherwise), the signal will be able to pass through later stages at a high level and so their noise contribution will be less significant. On the other hand, if the input stage is configured with a fixed gain, it will not be possible to turn it down to avoid clipping. Ideally the input stage should have variable gain. It is not straightforward to combine this feature with a balanced input, but several ways of doing it are shown later in this chapter.

Input Amplifier Functions

Firstly, RF filtering is applied at the very front end to prevent noise breakthrough and other EMC problems. It must be done before the incoming signal encounters any semiconductors where RF demodulation could occur, and can be regarded as a ‘roofing filter’. At the same time, the bandwidth at the low end is given an early limit by the use of DC-blocking capacitors, and in some cases overvoltage spikes are clamped by diodes. The input amplifier should present a reasonably high impedance to the outside world, not less than $10\text{ k}\Omega$, and preferably more. It must have a suitable gain – possibly switched or variable – to scale the incoming signal to the nominal internal level. Balanced input amplifiers also accurately perform the subtraction process that converts differential signals to single-ended ones, so noise produced by ground loops and the like is rejected. It’s quite a lot of work for one stage.

Unbalanced Inputs

The simplest unbalanced input feeds the incoming signal directly to the first stage of the audio chain. This is often impractical; for example, if the first stage was a Baxandall tone-control circuit then the boost and cut curves would be at the mercy of whatever source impedance was feeding the input. In addition, the input impedance would be low, and variable with frequency

and control settings. Some sort of buffer amplifier that can be fed from a significant impedance without ill effect is needed.

Figure 14.1 shows an unbalanced input amplifier, with the added components needed for interfacing to the real world. The op-amp U1:A acts as a unity-gain voltage-follower; it can be easily altered to give gain by adding two series feedback resistors. A 5532 bipolar type is used here for low noise; with the low source impedances that are likely to be encountered here, an FET-input op-amp would be 10 dB or more noisier. R1 and C1 are a first-order low-pass filter to remove incoming RF before it has a chance to reach the op-amp and demodulate into the audio band; once this has occurred any further attempts at RF filtering are, of course, pointless. R1 and C1 must be as close to the input socket as physically possible to prevent RF from being radiated inside the box before it is shunted to ground and so come before all other components in the signal path.

Selecting component values for input filters of this sort is always a compromise, because the output impedance of the source equipment is not known. If the source is an active preamplifier stage, then the output impedance will probably be around $50\ \Omega$, but it could be as high as $500\ \Omega$ or more. If the source is an oxymoronic ‘passive preamplifier’ – i.e. just an input selector switch and a volume potentiometer – then the output impedance will be a good deal higher. (At least one passive preamplifier uses a transformer with switched taps for volume control – see Chapter 9.) If you really want to use a piece of equipment that embodies its internal contradictions in its very name, then a reasonable potentiometer value is $10\ \text{k}\Omega$, and its maximum output impedance (when it is set for 6 dB of attenuation) will be $2.5\ \text{k}\Omega$, which is very different from the $50\ \Omega$ we might expect from a good active preamplifier. This is in series with R1 and affects the turnover frequency of the RF filter. Effective RF filtering is very desirable, but it is also important to avoid a frequency response that sags significantly at 20 kHz. Valve equipment is also likely to have a high output impedance.

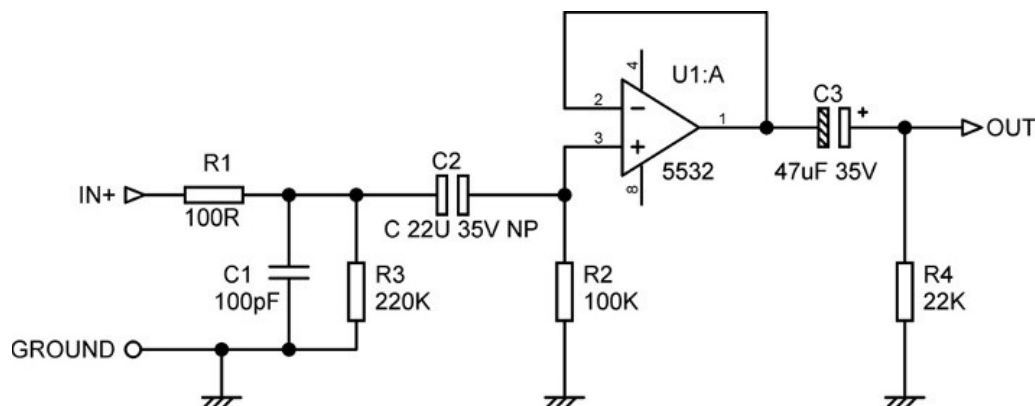


Figure 14.1: A typical unbalanced input amplifier with associated components

Taking $2.5\text{ k}\Omega$ as a worst-case source impedance and adding $R1$, then $2.6\text{ k}\Omega$ and 100 pF together give us -3 dB at 612 kHz ; this gives a 20 kHz loss of only 0.005 dB , so possibly $C1$ could be usefully increased; for example, if we made it 220 pF then the 20 kHz loss is still only 0.022 dB , but the -3 dB point is 278 kHz , much improving the rejection of what used to be called the Medium Wave. If we stick with $C1$ at 100 pF and assume an active output with a $50\text{ }\Omega$ impedance in the source equipment, then together with the $100\text{ }\Omega$ resistance of $R1$ the total is $150\text{ }\Omega$, which in conjunction with 100 pF gives us -3 dB at 10.6 MHz . This is rather higher than desirable, but it is not easy to see what to do about it, and we must accept the compromise. If there was a consensus that the output impedance of a respectable piece of audio equipment should not exceed $100\text{ }\Omega$, then things would be much easier.

Our compromise seems reasonable, but can we rely on $2.5\text{ k}\Omega$ as a worst-case source impedance? I did a quick survey of the potentiometer values that passive preamplifiers currently employ, and while it confirmed that $10\text{ k}\Omega$ seems to be the most popular value, one model had a $20\text{ k}\Omega$ potentiometer and another had a $100\text{ k}\Omega$ pot. The latter would have a maximum output impedance of $25\text{ k}\Omega$, and would give very different results with a $C1$ value of 100 pF – the worst-case frequency response would now be -3 dB at 63.4 kHz and -0.41 dB at 20 kHz , which is not helpful if you are aiming for a ruler-flat response in the audio band.

To put this into perspective, filter capacitor $C1$ will almost certainly be smaller than the capacitance of the interconnecting cable. Audio interconnect capacitance is usually in the range $50\text{--}150\text{ pF/meter}$, so with our assumed $2.5\text{ k}\Omega$ source impedance and 150 pF/meter cable, and ignoring $C1$, you can only permit yourself a rather short run of 3.3 meters before you are -0.1 dB down at 20 kHz , while with a $25\text{ k}\Omega$ source impedance you can hardly afford to have any cable at all; if you use low-capacitance 50 pF/meter cable you might just get away with a meter. This is just one of many reasons why ‘passive preamplifiers’ really are *not* a good idea.

Another important consideration is that the series resistance $R1$ must be kept as low as practicable to minimize Johnson noise, but lowering this resistance means increasing the value of shunt capacitor $C1$, and if it becomes too big then its impedance at high audio frequencies will become too low. Not only will there be too low a roll-off frequency if the source has a high output impedance, but there might be an increase in distortion at high audio frequencies because of excessive loading on the source output stage.

Replacing $R1$ with a small inductor to make an LC low-pass filter will give much better RF rejection at increased cost. This is justifiable in professional audio equipment, but it is much less common in hi-fi, one reason being that the unpredictable source impedance makes the filter design difficult, as we have just seen. In the professional world one *can* assume that the source impedance will be low. Adding more capacitors and inductors allows a three- or four-pole LC filter to be made. If you do use inductors then it is essential to check the frequency response to make sure it is what you expect and there is no peaking at the turnover frequency.

C2 is a DC-blocking capacitor to prevent voltages from ill-conceived source equipment getting into the circuitry. It is a non-polarized type as voltages from the outside world are of unpredictable polarity, and it is rated at not less than 35 V so that even if it gets connected to defective direct-coupled equipment with an op-amp output jammed hard against one of the supply rails, no harm will result. R3 is a DC drain resistor that prevents the charge put on C2 by the aforesaid external equipment from remaining there for a long time and causing a thud when connections are replugged; as with all input drain resistors, its value is a compromise between discharging the capacitor reasonably quickly and keeping the input impedance acceptably high. The input impedance here is R3 in parallel with R2, i.e. 220 k Ω in parallel with 100 k Ω , giving 68 k Ω . This is a good high value and should work well with just about any source equipment you can find, including valve technology.

R2 provides the biasing for the op-amp input; it must be a high value to keep the input impedance up, but bipolar input op-amps draw significant input bias current. The Fairchild 5532 data sheet quotes 200 nA typical and 800 nA maximum, and these currents would give a voltage drop across R2 of 20 and 80 mV respectively. This offset voltage will be reproduced at the output of the op-amp, with the input offset voltage added on; this is only 4 mV maximum and so will not affect the final voltage much, whatever its polarity. The 5532 has NPN input transistors, and the bias current flows into the input pins, so the voltage at pin 3 and hence the output will be negative with respect to ground by anything up to 84 mV.

Such offset voltages are not so great that the output voltage swing of the op-amp is significantly affected, but they are enough to generate unpleasant clicks and pops if the input stage is followed by any sort of switching, and enough to make potentiometers crackly. Output DC blocking is therefore required in the shape of C3, while R4 is another DC drain resistor to keep the output at zero volts. It can be made rather lower in value than the input drain resistor R3 as the only requirement is that it should not significantly load the op-amp output. FET-input op-amps have much lower input bias currents, so that the offsets they generate as they flow through biasing resistors are usually negligible, but they still have input offsets of a few millivolts, so DC blocking will still be needed if switches downstream are to work silently.

This input stage, with its input terminated by 50 Ω to ground, has a noise output of only -119.0 dBu over the usual 22 Hz – 22 kHz bandwidth. This is very quiet indeed, and is a reflection of the fact that R1, the only resistor in the signal path, has the low value of 100 Ω and so generates a very small amount of Johnson noise, only -132.6 dBu. This is swamped by the voltage noise of the op-amp, which is basically all we see; its current noise has negligible effect because of the low circuit impedances.

An unbalanced input stage with a 470 k Ω input impedance and gain variable over a 6 dB range for stereo balance control is described in Chapter 11.

Balanced Interconnections

Balanced inputs are used to prevent noise and crosstalk from affecting the input signal, especially in applications where long interconnections are used. They are standard on professional audio equipment, and are slowly but steadily becoming more common in the world of hi-fi. Their importance is that they can render ground loops and other connection imperfections harmless. Since there is no point in making a wonderful piece of equipment and then feeding it with an impaired signal, making an effective balanced input is of the first importance.

The basic principle of balanced interconnection is to get the signal you want by subtraction, using a three-wire connection. In some cases a balanced input is driven by a balanced output, with two anti-phase output signals – one signal wire (the hot or in-phase) sensing the in-phase output of the sending unit, while the other senses the anti-phase output.

In other cases, when a balanced input is driven by an unbalanced output, as shown in Figure 14.2, one signal wire (the hot or in-phase) senses the single output of the sending unit, while the other (the cold or phase-inverted) senses the unit's output-socket ground, and once again the difference between them gives the wanted signal. In either of these two cases, any noise voltages that appear identically on both lines (i.e. common-mode signals) are in theory completely canceled by the subtraction. In real life the subtraction falls short of perfection, as the gains via the hot and cold inputs will not be precisely the same, and the degree of discrimination actually achieved is called the common-mode rejection ratio (CMRR), of which more later.

It is deeply tedious to keep referring to non-inverting and inverting inputs, and so these are usually abbreviated to 'hot' and 'cold' respectively. This does *not* necessarily mean that the hot terminal carries more signal voltage than the cold one. For a true balanced connection, the voltages will be equal. The 'hot' and 'cold' terminals are also often referred to as In+ and In–, and this latter convention has been followed in the diagrams here.

The subject of balanced interconnections is a large one, and a big book could be written on this topic alone; one of the classic papers on the subject is by Muncy [1]. To make a start, let us look at the pros and cons of balanced connections.

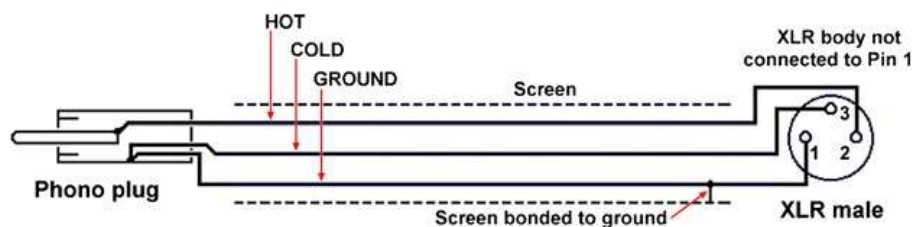


Figure 14.2: Unbalanced output to balanced input interconnection

The Advantages of Balanced Interconnections

- Balanced interconnections discriminate against noise and crosstalk, whether they result from ground currents, or electrostatic or magnetic coupling to signal conductors.
- Balanced connections make ground loops much less intrusive, and usually inaudible, so people are less tempted to start ‘lifting the ground’ to break the loop, with possibly fatal consequences. This tactic is only acceptable if the equipment has a dedicated ground-lift switch that leaves the external metalwork firmly connected to mains safety earth. In the absence of this switch the foolhardy and optimistic will break the mains earth (not quite so easy now that moulded mains plugs are standard) and this practice is of course highly dangerous, as a short-circuit from mains to the equipment chassis will result in live metalwork but dead people.
- A balanced interconnection incorporating a true balanced output gives 6 dB more signal level on the line, which should give 6 dB more dynamic range. However, this is true only with respect to *external* noise – as is described later in this chapter, a standard balanced input using 10 k Ω resistors is about 14 dB noisier than the unbalanced input shown in [Figure 14.1](#).
- Balanced connections are usually made with XLR connectors. These are a professional three-pin format, and are far superior to the phono (RCA) type normally used for unbalanced connections. More on this below.

The Disadvantages of Balanced Interconnections

- Balanced inputs are inherently noisier than unbalanced inputs by a large margin, in terms of the noise generated by the input circuitry itself rather than external noise. This may appear paradoxical but it is all too true, and the reasons will be fully explained in this chapter.
- More hardware means more cost. Small-signal electronics is relatively cheap; unless you are using a sophisticated low-noise input stage, of which more later, most of the extra cost is likely to be in the balanced input connectors.
- Balanced connections do not of themselves provide any greater RF immunity than an unbalanced input. For this to happen both legs of the balanced input would have to demodulate the RF in equal measure for common-mode cancellation to occur. This is highly unlikely, and the chances of it happening over a wide frequency range are zero. It remains vital to provide the usual passive RF filtering in front of any electronics to avoid EMC troubles.
- There is the possibility of introducing a phase error. It is all too easy to create an unwanted phase inversion by confusing hot and cold when wiring up a connector, and

this can go undiscovered for some time. The same mistake on an unbalanced system interrupts the audio completely and leaves no room for doubt.

Balanced Cables and Interference

In a balanced interconnection two wires carry the signal, and the third connection is the ground wire, which has two functions. Firstly, it joins the grounds of the interconnected equipment together. This is not always desirable, and if galvanic isolation is required a transformer balancing system will be necessary because the large common-mode voltages are likely to exceed the range of an electronic balanced input. A good transformer will also have a very high CMRR, which will be needed to get a clean signal in the face of large CM voltages.

Secondly, the presence of the ground allows electrostatic screening of the two signal wires, preventing both the emission and pickup of unwanted signals. This can mean:

1. a lapped screen, with wires laid parallel to the central signal conductor. The screening coverage is not total, and can be badly degraded as the screen tends to open up on the outside of cable bends. Not recommended unless cost is the dominating factor;
2. a braided screen around the central signal wires. This is much more expensive, as it is harder to make, but opens up less on bending than a lap screen. Even so, screening is not 100%. It has to be said that it is a pain to terminate in the usual audio connectors. Not recommended;
3. an overlapping foil screen, with the ground wire (called the drain wire in this context for some reason) running down the inside of the foil and in electrical contact with it. This is usually the most effective as the foil is a solid sheet and cannot open up on the outside of bends. It should give perfect electrostatic screening, and it is much easier to work with than either lap screen or braided cable. However, the higher resistance of aluminum foil compared with copper braid means that RF immunity may not be so good.

There are three main ways in which an interconnection is susceptible to hum and noise.

1 *Electrostatic Coupling*

An interfering signal at significant voltage couples directly to the inner signal line, through stray capacitance. The stray capacitance between imperfectly screened conductors will be a fraction of a pF in most circumstances, as electrostatic coupling falls off with the square of distance. This form of coupling can be serious in studio installations with unrelated signals running down the same ducting.

The three main lines of defence against electrostatic coupling are effective screening, low impedance drive, and a good CMRR maintained up to the top of the audio spectrum. As regards screening, an overlapped foil screen provides complete protection.

Driving the line from a low impedance, of the order of $100\ \Omega$ or less, is also helpful because the interfering signal, having passed through a very small stray capacitance, is a very small current and cannot develop much voltage across such a low impedance. This is convenient because there are other reasons for using a low output impedance, such as optimizing the interconnection CMRR, minimizing HF losses due to cable capacitance, and driving multiple inputs without introducing gain errors. For the best immunity to crosstalk the output impedance must remain low up to as high a frequency as possible. This is definitely an issue as op-amps invariably have a feedback factor that begins to fall from a low, and quite possibly sub-audio, frequency and this makes the output impedance rise with frequency as the negative feedback factor falls, as if an inductor were in series. Some line outputs have physical series inductors to improve stability or EMC immunity, and these should not be so large that they significantly increase the output impedance at 20 kHz. From the point of view of electrostatic screening alone, the screen does not need to be grounded at both ends, or form part of a circuit [2]. It must of course be grounded at some point.

If the screening is imperfect, and the line impedance non-zero, some of the interfering signal will get into the hot and cold conductors, and now the CMRR must be relied upon to make the immunity acceptable. If it is possible, rearranging the cable-run away from the source of interference and getting some properly screened cable is more practical and more cost-effective than relying on very good common-mode rejection.

Stereo hi-fi balanced interconnections almost invariably use XLR connectors. Since the usual three-pin XLR can only handle one balanced channel, two separate cables are almost invariably used and interchannel capacitive crosstalk is not an issue. Professional systems, on the other hand, use multi-way connectors that do not have screening between the pins and there is an opportunity for capacitive crosstalk here, but the use of low source impedances should reduce it to below the noise floor.

2 Magnetic Coupling

If a cable runs through an AC magnetic field, an EMF is induced both in signal conductors and in the screen, and according to some writers the screen current must be allowed to flow freely or its magnetic field will not cancel out the field acting on the signal conductors, and therefore the screen should be grounded at both ends, to form a circuit [3]. In practice the magnetic field cancellation will be very imperfect and reliance is better placed on the CMRR of the balanced system to cancel out the hopefully equal voltages induced in the two signal wires. The need to ground both ends to possibly optimize the magnetic rejection is not usually a restriction, as it is rare that galvanic isolation is required between two pieces of audio equipment.

The equality of the induced voltages can be maximized by minimizing the loop area between the hot and cold signal wires, for example by twisting them tightly together in manufacture. In practice most audio foil-screen cables have parallel rather than twisted signal conductors,

but this seems adequate almost all of the time. Magnetic coupling falls off with the square of distance, so rearranging the cable-run away from the source of magnetic field is usually all that is required. It is unusual for it to present serious difficulties in a hi-fi application.

3 Ground Voltages

These are the result of current flowing through the ground connection, and is often called ‘common-impedance coupling’ in the literature [1]. This is the root of most ground-loop problems. The existence of a loop in itself does no harm, but it is invariably immersed in a 50 Hz magnetic field that induces mains-frequency currents plus harmonics into it. This current produces a voltage drop down non-negligible ground-wire resistances, and this effectively appears as a voltage source in each of the two signal lines. Since the CMRR is finite a proportion of this voltage will appear to be a differential signal, and will be reproduced as such.

Balanced Connectors

Balanced connections are most commonly made with XLR connectors, though it can be done with stereo (tip-ring-sleeve) jack plugs. XLRs are a professional three-pin format, and are a much better connector in every way than the usual phono (RCA) connectors used for unbalanced interconnections. Phono connectors have the great disadvantage that if you are connecting them with the system active (inadvisable, but then people are always doing inadvisable things) the signal contacts meet before the grounds and thunderous noises result. The XLR standard has pin 2 as hot, pin 3 as cold, and pin 1 as ground.

Stereo jack plugs are often used for line-level signals in a recording environment, and are frequently found on the rear of professional power amplifiers as an alternative to an adjacent XLR connector; both full-size and 3.5 mm sizes are used. Balanced jacks are wired with the tip as hot, the ring as cold, and the sleeve as ground. Sound reinforcement systems often use large multi-way connectors that carry dozens of three-wire balanced connections.

Balanced Signal Levels

Many pieces of equipment, including preamplifiers and power amplifiers designed to work together, have both unbalanced and balanced inputs and outputs. The general consensus in the hi-fi world is that if the unbalanced output is say 1 Vrms, then the balanced output will be created by feeding the in-phase output to the hot output pin, and also to a unity-gain inverting stage, which drives the cold output pin with 1 Vrms phase-inverted. The total balanced output voltage between hot and cold pins is therefore 2 Vrms, and so the balanced input must have a gain of 0.5 or -6 dB relative to the unbalanced input to maintain consistent internal signal levels.

Electronic vs. Transformer Balanced Inputs

Balanced interconnections can be made using either transformer or electronic balancing. Electronic balancing has many advantages, such as low cost, low size and weight, superior frequency and transient response, and no low-frequency linearity problems. It may still be regarded as a second-best solution in some quarters, but the performance is more than adequate for most professional applications. Transformer balancing does have some advantages of its own, particularly for work in very hostile RF/EMC environments, but serious drawbacks. The advantages are that transformers are electrically bulletproof, retain their high CMRR performance forever, and consume no power even at high signal levels. They are essential if galvanic isolation between ground is required. Unfortunately transformers generate LF distortion, particularly if they have been made with minimal core sizes to save weight and cost. They are liable to have HF response problems due to leakage reactance and distributed capacitance, and compensating for this requires a carefully designed Zobel network across the secondary. Inevitably they are heavy and expensive. Transformer balancing is therefore relatively rare, even in professional audio applications, and the greater part of this chapter deals with electronically balanced inputs.

Common-Mode Rejection

Figure 14.3 shows a balanced interconnection reduced to its bare essentials: hot and cold line outputs with source resistances R_{out+} , R_{out-} and a standard differential amplifier at the input end. The output resistances are assumed to be exactly equal, and the balanced input in the receiving equipment has two exactly equal input resistances to ground $R1$, $R2$. The ideal balanced input amplifier senses the voltage difference between the points marked $In+$ (hot) and $In-$ (cold), and ignores any common-mode voltages that are present on both. The amount by which it discriminates is called the common-mode rejection ratio or CMRR, and is usually measured in dB. Suppose a differential voltage input between $In+$ and $In-$ gives an output

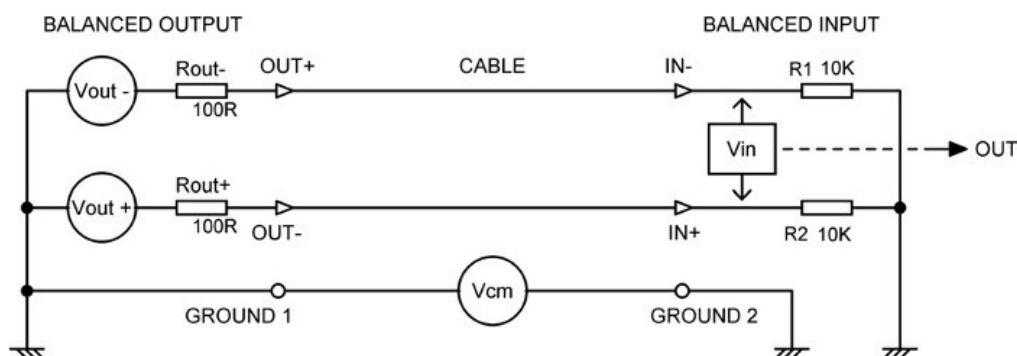


Figure 14.3: A theoretical balanced interconnection showing how the output and input impedances influence CMRR

voltage of 0 dB; now reconnect the input so that $In+$ and $In-$ are joined together and the same voltage is applied between them and ground. Ideally the result would be zero output, but in this imperfect world it won't be, and the output could be anywhere between -20 dB (for a bad balanced interconnection, which probably has something wrong with it) and -140 dB (for an extremely good one). The CMRR when plotted may have a flat section at low frequencies, but it very commonly degrades at high audio frequencies and may also deteriorate at very low frequencies. More on that later.

In one respect balanced audio connections have it easy. The common-mode signal is normally well below the level of the unwanted signal, and so the common-mode range of the input is not an issue. In other areas of technology, such as electrocardiogram amplifiers, the common-mode signal may be many times greater than the wanted signal.

The simplified conceptual circuit of [Figure 14.3](#), under SPICE simulation, demonstrates the need to get the resistor values right for a good CMRR, before you even begin to consider the rest of the circuitry. The differential voltage sources V_{out+} , V_{out-} representing the actual balanced output are set to zero, and V_{cm} , which represents the common-mode voltage drop down the cable ground, is set to 1 V to give a convenient result in dBV. The output resulting from the presence of this voltage source is measured by a mathematical subtraction of the voltages at $In+$ and $In-$ so there is no actual input amplifier to confuse the results with its non-ideal performance.

Let us begin with R_{out+} and R_{out-} set to $100\ \Omega$ and $R1$ and $R2$ set to $10\ k\Omega$. These are typical real-life values as well as being nice round figures. When all four resistances are exactly at their nominal value, the CMRR is in theory infinite, which my SPICE simulator rather curiously reports as exactly -400 dB. If one of the output resistors or one of the input resistors is then altered in value by 1%, then the CMRR drops like a stone to -80 dB. If the deviation from equality is 10%, things are predictably worse and the CMRR degrades to -60 dB, as shown in [Table 14.2](#). That would be quite a good figure in reality, but since we have not yet even thought about op-amp imperfections or other circuit imbalances, and have only altered one resistance out of the four that will in real circuitry all have their own tolerances, it underlines the need to get things right at the most basic theoretical level before we dig deeper into the circuitry. The CMRR is naturally flat with frequency because our simple model has no frequency-dependent components.

The essence of the problem is that we have two resistive dividers, and to get an infinite CMRR they must have exactly the same attenuation. If we increase the ratio between the output and input resistors, by reducing the former or increasing the latter, the attenuation factor becomes closer to unity, so variations in either resistor value have less effect on it. If we increase the input impedance to $100\ k\Omega$, which is quite practical in real life (we will put aside the noise implications of this for the moment), things are 10 times better, as the R_{in}/R_{out} ratio has improved from 100 to 1000 times. We now get a CMRR of -100 dB with a 1% resistance deviation, and -80 dB with a 10% deviation. An even higher input impedance of $1\ M\Omega$, which

TABLE 14.2 How resistor tolerances affect the theoretical CMRR of the theoretical circuit in Figure 14.3

R_{out+}	R_{out-}	R_{out} deviation	R1	R2	R1, R2 deviation	R_{in}/R_{out} ratio	CMRR (dB)
100	100	0	10k	10k	0	100	Infinity
100	101	1%	10k	10k	0	100	-80.2
100	110	10%	10k	10k	0	100	-60.2
100	100	0	10k	10.1k	1%	100	-80.3
100	100	0	10k	11k	10%	100	-61.0
100	100	0	100k	101k	1%	1000	-100.1
100	100	0	100k	110k	10%	1000	-80.8
100	100	0	1M	1.01M	1%	10,000	-120.1
100	100	0	1M	1.1M	10%	10,000	-100.8
68	68	0	20k	20.2k	1%	294	-89.5
68	68	0	20k	22k	10%	294	-70.3

is perhaps a bit less practical, raises R_{in}/R_{out} to 10,000 and gives -120 dB for a 1% resistance deviation and -100 dB for a 10% deviation.

We can attack the other aspect of the attenuation problem by reducing the output impedances to 10 Ω , ignoring for the moment the need to secure against HF instability caused by cable capacitance, and also return the input impedance resistors to 100 k Ω . R_{in}/R_{out} is 10,000 once more, and, as you might suspect, the CMRR is once more -120 dB for a 1% deviation and -100 dB for a 10% deviation. Ways to make stable output stages with very low output impedances are described in Chapter 15; a fraction of an ohm at 1 kHz is quite easy to achieve.

In conventional circuits, the combination of 68 Ω output resistors and a 20 k Ω input impedance is often encountered, 68 Ω being about as low as you want to go if HF instability is to be absolutely guarded against with the long lines used in professional audio. The 20 k Ω common-mode input impedance is what you get if you make a basic balanced input amplifier with four 10 k Ω resistors. I strongly suspect that this value is so popular because it looks as if it gives standard 10 k Ω input impedances; in fact it does nothing of the sort, and the common-mode input impedance, which is what matters here, is 20 k Ω on each leg – more on that later. It turns out that 68 Ω output resistors and a 20 k Ω input impedance give a theoretical CMRR of -89.5 dB for a 1% deviation of one resistor, which is quite encouraging. These results are summarized in Table 14.2.

The conclusion is simple: we need the lowest possible output impedances and the highest possible input impedances to get the maximum common-mode rejection. This is highly

convenient because low output impedances are already needed to drive multiple amplifier inputs and cable capacitance, and high input impedances are needed to minimize loading and maximize the number of amplifiers that can be driven.

The Basic Electronic Balanced Input

Figure 14.4 shows the basic balanced input amplifier using a single op-amp. To achieve balance $R1$ must be equal to $R3$ and $R2$ equal to $R4$. The amplifier in Figure 14.5 has a gain of $R2/R1$ ($= R4/R3$). The standard one-op-amp balanced input or differential amplifier is a very familiar circuit block, but its operation often appears somewhat mysterious. Its input impedances are *not* equal when it is driven from a balanced output; this has often been commented on [4], and some confusion has resulted.

The source of the confusion is that a simple differential amplifier has interaction between the two inputs, so that the input impedance seen on the cold input depends on the signal applied to the hot input. Input impedance is measured by applying a signal and seeing how much current flows into the input, so it follows that the apparent input impedance on each leg varies

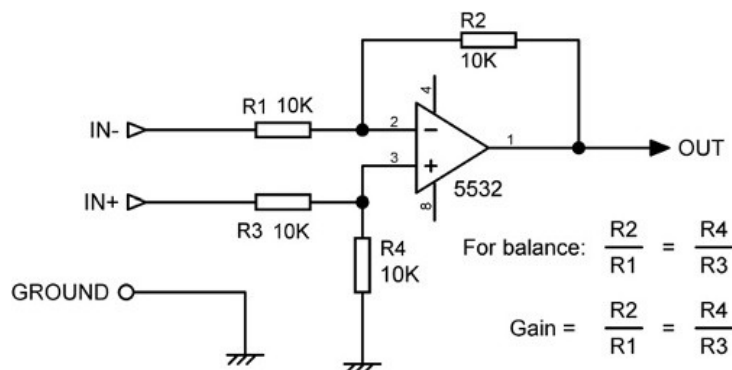


Figure 14.4: The basic balanced input amplifier

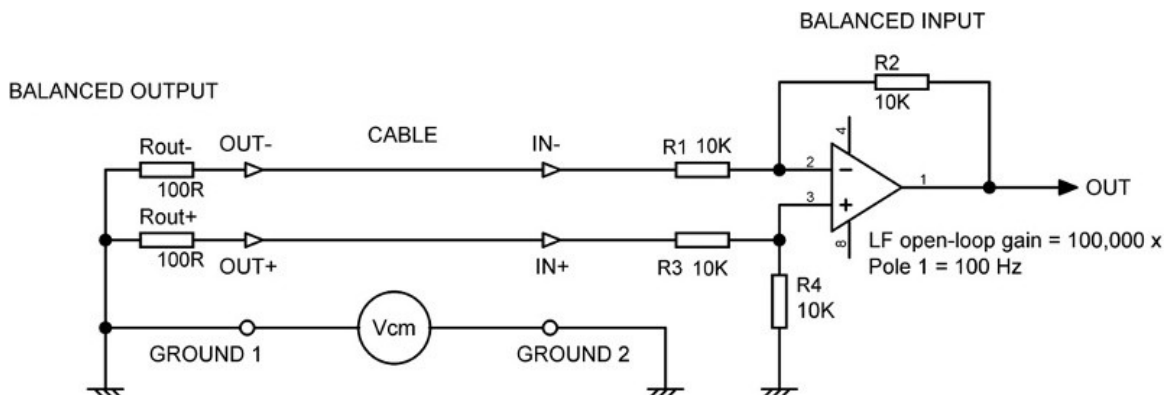


Figure 14.5: A simple balanced interconnection for SPICE simulation to show the effect that op-amp properties have on the CMRR

according to how the cold input is driven. If the amplifier is made with four $10\text{ k}\Omega$ resistors, then the input impedances on hot and cold are as shown in Table 14.3.

Some of these impedances are not exactly what you might expect, and require some explanation.

Case 1. The balanced input is being used as an unbalanced input by grounding the cold input and driving the hot input only. The input impedance is therefore simply $R3 + R4$. Resistors $R3$ and $R4$ reduce the signal by a factor of a half, but this loss is undone as $R1$ and $R2$ set the amplifier gain to 2 times, and the overall gain is unity. If the cold input is not grounded then the gain is 0.5 times. The attenuate-then-amplify architecture, plus the Johnson noise from the resistors, makes this configuration much noisier than the dedicated unbalanced input of Figure 14.1, which has only a single $100\text{ }\Omega$ resistor in the signal path.

Case 2. The balanced input is again being used as an unbalanced input, but this time by grounding the hot input, and driving the cold input only. This gives a phase inversion and it is unlikely you would want to do it except as an emergency measure to correct a phase error somewhere else. The important point here is that the input impedance is now only $10\text{ k}\Omega$, the value of $R1$, because shunt negative feedback through $R2$ creates a virtual earth at pin 2 of the op-amp. Clearly this simple circuit is not as symmetrical as it looks. The gain is unity, whether or not the hot input is grounded; grounding it is desirable because it not only prevents interference being picked up on the hot input pin, but also puts $R3$ and $R4$ in parallel, reducing the resistance from op-amp pin 3 to ground and so reducing Johnson noise.

Case 3. This is the standard balanced interconnection. The input is driven from a balanced output with the same signal levels on hot and cold, as if from a transformer with its center-tap grounded, or an electronically balanced output using a simple inverter to drive the cold pin. The input impedance on the hot input is what you would expect; $R3 + R4$ add up to $20\text{ k}\Omega$. However, on the cold input there is a much lower input impedance of $6.66\text{ k}\Omega$. This at first sounds impossible as the first thing the signal encounters is a $10\text{ k}\Omega$ series resistor, but the crucial point is that the hot input is being driven simultaneously with a signal of the opposite phase, so the inverting op-amp input is moving in the opposite direction to the cold input due to negative feedback, and what you might call anti-bootstrapping reduces the effective value of the $10\text{ k}\Omega$ resistor to $6.66\text{ k}\Omega$. These are the differential input impedances we are examining,

TABLE 14.3 The input impedances for different input drive conditions

Case	Pins driven	Hot input res. (Ω)	Cold input res. (Ω)
1	Hot only	20k	Grounded
2	Cold only	Grounded	10k
3	Both (balanced)	20k	6.66k
4	Both common-mode	20k	20k
5	Both floating	10k	10k

the impedances seen by the balanced output driving them. Common-mode signals see a common-mode impedance of $20\text{ k}\Omega$, as in Case 4 below.

You will sometimes see the statement that these unequal differential input impedances ‘unbalance the line’. From the point of view of CMRR, this is not the case, as it is the CM input impedance that counts. The line is, however, unbalanced in the sense that the cold input draws three times the current from the output that the hot one does. This current imbalance might conceivably lead to inductive crosstalk in some multi-way cable situations, but I have never encountered it. The differential input impedances can be made equal by increasing the $R1$ and $R2$ resistor values by a factor of 3, but this degrades the noise performance markedly and makes the common-mode impedances to ground unequal, which is a much worse situation as it compromises the rejection of ground voltages, and these are almost always the main problem in real life.

Case 4. Here both inputs are driven by the same signal, representing the existence of a common-mode voltage. Now both inputs shown an impedance of $20\text{ k}\Omega$. It is the symmetry of the common-mode input impedances that determines how effectively the balanced input rejects the common-mode signal. This configuration is of course only used for CMRR testing.

Case 5. Now the input is driven as from a floating transformer with the center-tap (if any) unconnected, and the impedances can be regarded as equal; they must be, because with a floating winding the same current must flow into each input. However, in this connection the line voltages are *not* equal and opposite: with a true floating transformer winding the hot input has all the signal voltage on it while the cold has none at all, due to the negative feedback action of the balanced input amplifier. This seemed very strange when it emerged in SPICE simulation, but a sanity check with real components proves it true. The line has been completely unbalanced as regards crosstalk to other lines, although its own common-mode rejection remains good.

Even if absolutely accurate resistors are assumed, the CMRR of the stage in [Figure 14.4](#) is not infinite; with a TL072 it is about -90 dB , degrading from 100 Hz upwards, due to the limited open-loop gain of the op-amp. We will now examine this effect.

Common-Mode Rejection: The Basic Balanced Input and Op-Amp Effects

In the earlier section on CMRR we saw that in a theoretical balanced line, choosing low output impedances and high input impedances would give very good CM rejection even if the resistors were not perfectly matched. Things are a bit more complex (i.e. worse) if we replace the mathematical subtraction with a real op-amp. We quickly find that even if perfectly matched resistors everywhere are assumed, the CMRR of the stage is not infinite, because the two op-amp inputs are not at exactly the same voltage. The negative feedback error voltage between the inputs depends on the open-loop gain of the op-amp, and that is neither infinite nor

flat with frequency into the far ultraviolet. Far from it. There is also the fact that op-amps themselves have a CMRR; it is high, but once more it is not infinite.

As usual, SPICE simulation is instructive, and [Figure 14.5](#) shows a simple balanced interconnection, with the balanced output represented simply by two $100\ \Omega$ output resistances connected to the source equipment ground, here called Ground 1, and the usual differential op-amp configuration at the input end, where we have Ground 2.

A common-mode voltage V_{cm} is now injected between Ground 1 and Ground 2, and the signal between the op-amp output and Ground 2 measured. The balanced input amplifier has all four of its resistances set to precisely $10\ \text{k}\Omega$, and the op-amp is represented by a very simple model that has only two parameters: a low-frequency open-loop gain, and a single pole frequency that says where that gain begins to roll-off at 6 dB/octave. The op-amp input impedances and the op-amp's own CMRR are assumed infinite, as in the world of simulation they so easily can be. Its output impedance is set at zero.

For the first experiments, even the pole frequency is made infinite, so now the only contact with harsh reality is that the op-amp open-loop gain is finite. That is, however, enough to give distinctly non-ideal CMRR figures, as [Table 14.4](#) shows.

With a low-frequency open-loop gain of 100,000, which happens to be the typical figure for a 5532 op-amp, even perfect components everywhere will never yield a better CMRR than $-94\ \text{dB}$. The CMRR is shown as a raw ratio in the third column so you can see that the CMRR is inversely proportional to the gain, and so we want as much gain as possible.

Op-Amp Frequency Response Effects

To examine these we will set the low-frequency gain to 100,000, which gives a CMRR 'floor' of $-94\ \text{dB}$, and then introduce the pole frequency that determines where it rolls off. The CMRR now worsens at 6 dB/octave, starting at a frequency set by the interaction of the low-frequency gain and the pole frequency. The results are summarized in [Table 14.5](#), which shows that, as you might expect, the lower the open-loop bandwidth of the op-amp, the lower the frequency at which the CMRR begins to fall off. [Figure 14.6](#) shows the situation diagrammatically.

TABLE 14.4 The effect of finite op-amp gain on CMRR for the circuit of [Figure 14.5](#)

Open-loop gain	CMRR (dB)	CMRR ratio
10,000	-74.0	19.9×10^{-5}
30,000	-83.6	66.4×10^{-6}
100,000	-94.0	19.9×10^{-6}
300,000	-103.6	6.64×10^{-6}
1,000,000	-114.1	1.97×10^{-6}

Table 14.6 gives the open-loop gain and pole parameters for a few op-amps of interest. Both parameters, but especially the gain, are subject to considerable variation; the typical values from the manufacturers' data sheets are given here.

Some of these op-amps have very high open-loop gains, but only at very low frequencies. This may be good for DC applications, but in audio line-input applications, where the lowest frequency of CMRR interest is 50 Hz, they will be operating above the pole frequency and so the gain available will be less – possibly considerably so, in the case of op-amps like the OPA2134. This is not, however, a real limitation, for even if a humble TL072 is used the

TABLE 14.5 The effect of op-amp open-loop pole frequency on CMRR for the circuit of Figure 14.5

Pole frequency	CMRR break-point frequency
10 kHz	10.2 kHz
1 kHz	1.02 kHz
100 Hz	102 Hz
10 Hz	10.2 Hz

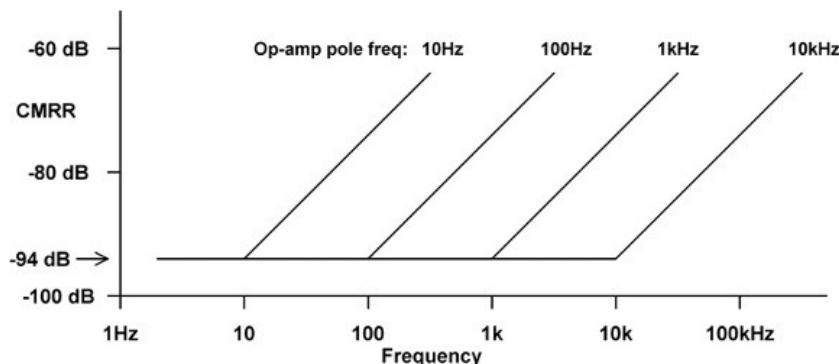


Figure 14.6: How the CMRR degrades with frequency for different op-amp pole frequencies. All resistors are assumed to be perfectly matched

TABLE 14.6 Typical LF gain and open-loop pole frequency for some op-amps commonly used in audio

Name	Input device type	LF gain	Pole frequency (Hz)	Op-amp LF CMRR (dB)
NE5532	Bipolar	100,000	100	100
LM4562	Bipolar	10,000,000	Below 10	120
LT1028	Bipolar	20,000,000	3	120
TL072	FET	200,000	20	86
OP27	FET	1,800,000	3	120
OPA2134	FET	1,000,000	3	100
OPA627	FET	1,000,000	20	116

perfect-resistor CMRR is about -90 dB, degrading from 100 Hz upwards. This sort of performance is not attainable in practice. We will shortly see why not.

Op-Amp CMRR Effects

Op-amps have their own CMRR, and we need to know how much this will affect the final CMRR of the balanced interconnection. The answer is that if all resistors are accurate, the overall CMRR is equal to the CMRR of the op-amp [5]. Since op-amp CMRR is typically very high (see the examples in Table 14.6), it is very unlikely to be the limiting factor.

The CMRR of an op-amp begins to degrade above a certain frequency, typically at 6 dB/octave. This is (fortunately) at a higher frequency than the open-loop pole, and is frequently around 1 kHz. For example, the OP27 has a pole frequency at about 3 Hz, but the CMRR remains flat at 120 dB until 2 kHz, and it is still greater than 100 dB at 20 kHz.

Amplifier Component Mismatch Effects

We saw earlier in this chapter that when the output and input impedances on a balanced line have a high ratio between them and are accurately matched we got a very good CMRR; this was compromised by the imperfections of op-amps, but the overall results were still very good – and much higher than the CMRRs measured in practice. There remains one place where we are still away in theory-land: we have so far assumed the resistances around the op-amp were all exactly accurate. We must now face reality, admit that these resistors will not be perfect, and see how much damage to the CMRR they will do.

SPICE simulation gives us Table 14.7. The situation with LF op-amp gains of both 100,000 and 1,000,000 is examined, but the effects of finite op-amp bandwidth or op-amp CMRR are not included. R1 in Figure 14.5 is varied while R2, R3, and R4 are all kept at precisely 10 k Ω , and the balanced output source impedances are set to exactly 100 Ω .

Table 14.7 shows with glaring clarity that our previous investigations, which took only output and input impedances into account, and determined that 68 Ω output resistors and 20 k Ω input impedances gave a CMRR of -89.5 dB for a 1% deviation in either, were actually quite unrealistic, and even adding in op-amp imperfections left us with unduly optimistic results. If a 1% tolerance resistor is used for R1 (and nowadays there is no financial incentive to use anything less accurate), the CMRR is dragged down at once to -46 dB; the same figure results from varying any other one of the four resistances by itself. If you are prepared to shell out for 0.1% tolerance resistors, the CMRR is a rather better -66 dB.

This shows that there really is no point in worrying about the gain of the op-amp you use in balanced inputs; the effect of mismatches in the resistors around that op-amp are far greater.

TABLE 14.7 How resistor tolerances affect the CMRR with some realistic op-amp open-loop gains

R1 (Ω)	R1 deviation (%)	Gain ($\times$)	CMRR (dB)
10k	0	100,000	−94.0
10.001k	0.01	100,000	−90.6
10.01k	0.1	100,000	−66.5
10.1k	1	100,000	−46.2
11k	10	100,000	−26.6
10k	0	1,000,000	−114.1
10.001k	0.01	1,000,000	−86.5
10.01k	0.1	1,000,000	−66.2
10.1k	1	1,000,000	−46.2
11k	10	1,000,000	−26.6

The results in the table give an illustration of how resistor accuracy affects CMRR, but it is only an illustration, because in real life – a phrase that seems to keep cropping up, showing how many factors affect a practical balanced interconnection – all four resistors will of course be subject to a tolerance, and a more realistic calculation would produce a statistical distribution of CMRR rather than a single figure. One method is to use the Monte Carlo function in SPICE, which runs multiple simulations with random component variations and collates the results. However you do it you must know (or assume) how the resistor values are distributed with their tolerance window. Usually you don't know, and finding out by measuring hundreds of resistors is not a task that appeals to all of us.

It is straightforward to assess the worst-case CMRR, which occurs when all resistors are at the limit of the tolerance in the most unfavorable direction. The CMRR in dB is then:

$$\text{CMRR} = 20 \log \left(\frac{1 + R2/R1}{4T/100} \right) \quad (\text{Equation 14.1})$$

where R1 and R2 are as in [Figure 14.5](#), and T is the tolerance (%).

This deeply pessimistic equation tells us that 1% resistors give a worst-case CMRR of only 34.0 dB, that 0.5% parts give only 40.0 dB and expensive 0.1% parts yield but 54.0 dB. Things are not, however, quite that bad in actuality, as the chance of everything being as wrong as possible is actually very small. I have measured the CMRR of more of these balanced inputs, built with 1% resistors, than I care to contemplate, but I do not recall that I ever saw one with an LF CMRR worse than 40 dB.

There are eight-pin SIL packages that offer four resistors that ought to have good matching, if not accurate absolute values; be very, very wary of these as they usually contain thick-film

resistive elements that are not perfectly linear. In a test I did a 10 k Ω SIL resistor with 10 Vrms across it generated 0.0010% distortion. Not a huge amount perhaps, but in the quest for perfect audio, resistors that do not stick to Ohm's Law are not a good start.

To conclude this section, it is clear that in practical use it is the errors in the balanced amplifier resistors that determine the CMRR, though both unbalanced capacitances (C1, C2 in Figure 14.7) and the finite op-amp bandwidth are likely to cause further degradation at high audio frequencies. If you are designing both ends of a balanced interconnection and you are spending money on precision resistors, you should put them in the input amplifier, not the balanced output. The LF gain of the op-amp, and op-amp CMRR, have virtually no effect.

At this point it has to be said that balanced input amplifiers with four 1% resistors around them are used extensively in the audio business, and almost always prove to have adequate CMRR. When more CMRR is thought desirable, for example in high-end mixing consoles, one of the resistances is made trimmable with a preset. This means a lot of tweaking in manufacture, but the upside is that this is a quick set-and-forget adjustment that will not need to be touched again unless one of the four resistors needs replacing, and that is unlikely. CMRRs at LF of more than 60 dB can easily be obtained by this method, but the CMRR at HF will degrade due to the op-amp gain roll-off.

A Practical Balanced Input

The simple balanced input circuit shown in Figure 14.4 is not fit to face the outside world without additional components. Figure 14.7 shows a fully equipped version. Firstly, and most important,

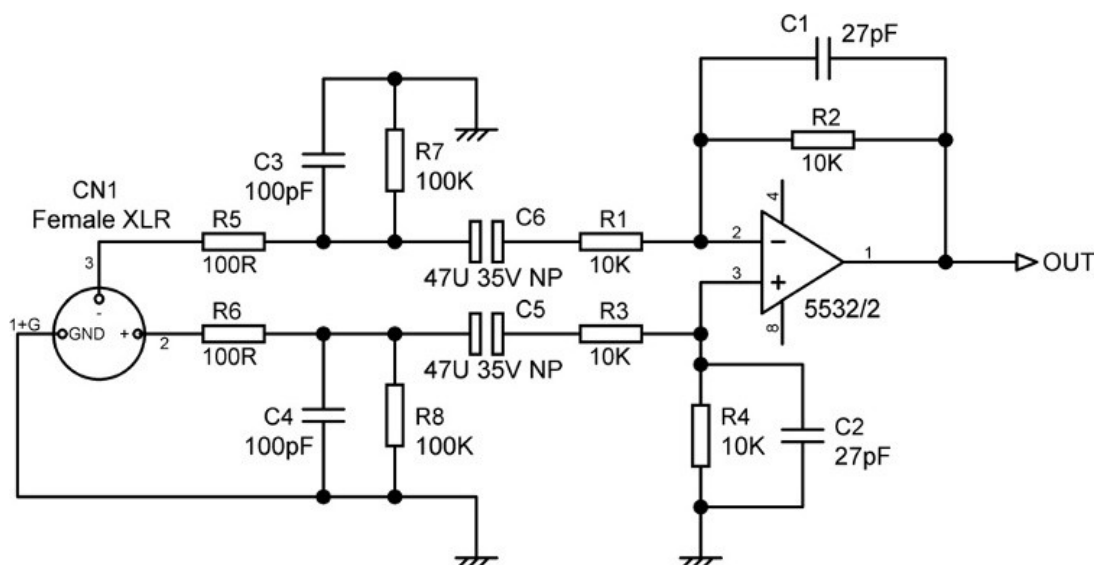


Figure 14.7: Balanced input amplifier with the extra components required for DC blocking and EMC immunity

C1 has been added across the feedback resistor R2; this prevents stray capacitances from pin 2 to ground causing extra phase-shifts that lead to HF instability. The value required for stability is small, much less than that which would cause an HF roll-off anywhere near the top of the audio band. The values here of 10k and 27 pF give -3 dB at 589 kHz, and such a roll-off is only down by 0.005 dB at 20 kHz. C2, of equal value, must be added across R4 to maintain the balance of the amplifier, and hence its CMRR, at high frequencies.

C1 and C2 must not be relied upon for EMC immunity as C1 is not connected to ground, and there is every chance that RF will demodulate at the op-amp inputs. A passive RF filter is therefore added to each input, in the shape of R5, C3 and R6, C4, so the capacitors will shunt incoming RF to ground before it reaches the op-amp. Put these as close to the input socket as possible to minimize radiation inside the enclosure.

I explained earlier in this chapter when looking at unbalanced inputs that it is not easy to guess what the maximum source impedance will be, given the existence of ‘passive preamplifiers’ and valve equipment. Neither are likely to have a balanced output, unless implemented by transformer, but either might be used to feed a balanced input, and so the matter needs thought.

In the unbalanced input, circuit resistances had to be kept as low as practicable to minimize the generation of Johnson noise that would compromise the inherently low noise of the stage. The situation with a standard balanced input is, however, different from the unbalanced case as there have to be resistances around the op-amp, and they must be kept up to a certain value to give acceptably high input impedances; this is why a balanced input like this one is much noisier. We could therefore make R5 and R6 much larger without a measurable noise penalty if we reduce R1 and R3 accordingly to keep unity gain. In [Figure 14.7](#) R5 and R6 are kept at 100 Ω , so if we assume 50 Ω output resistances in both legs of the source equipment, then we have a total of 150 Ω , and 150 Ω and 100 pF give -3 dB at 10.6 MHz. Returning to a possible passive preamplifier with a 10 k Ω potentiometer, its maximum output impedance of 2.5k plus 100 Ω with 100 pF gives -3 dB at 612 kHz, which remains well clear of the top of the audio band.

As with the unbalanced input, replacing R5 and R6 with small inductors will give much better RF filtering but at increased cost. Ideally a common-mode choke (two bifilar windings on a small toroidal core) should be used as this improves performance. Check the frequency response to make sure the LC circuits are well damped and not peaking at the turnover frequency.

C5 and C6 are DC-blocking capacitors. They must be rated at no less than 35 V to protect the input circuitry, and are the non-polarized type as external voltages are of unpredictable polarity. The lowest input impedance that can occur with this circuit when using 10 k Ω resistors is, as described above, 6.66 k Ω when it is being driven in the balanced mode. The low-frequency roll-off is therefore -3 dB at 0.51 Hz. This may appear to be undesirably low, but the important point is not the LF roll-off but the possible loss of CMRR at low frequencies due to imbalance in the values of C5 and C6; they are electrolytics with a significant

tolerance. Therefore they should be made large so their impedance is a small part of the total input impedance; 47 μF is shown here but 100 or 220 μF can be used to advantage if there is the space to fit them in. The low-end frequency response must be defined somewhere in the input system, and the earlier the better, to prevent headroom or linearity being affected by subsonic disturbances, but this is not a good place to do it. A suitable time-constant immediately after the input amplifier is the way to go, but remember that capacitors used as time-constants may distort unless they are NP0 ceramic, polystyrene, or polypropylene (see Chapter 2 for more on this).

R7, R8 are DC drain resistors to prevent charges lingering on C5 and C6. These can be made lower than for the unbalanced input as the input impedances are lower, so a value of, say, 100 $\text{k}\Omega$ rather than 220 $\text{k}\Omega$ makes relatively little difference to the total input impedance.

A useful property of this kind of balanced amplifier is that it does not go mad when the inputs are left open-circuit – in fact it is actually *less* noisy than with its inputs shorted to ground. This is the opposite of the ‘normal’ behavior of a high-impedance unterminated input. This is because two things happen: open-circuiting the hot input doubles the resistance seen by the non-inverting input of the op-amp, raising its noise contribution by 3 dB. However, opening the cold input makes the noise gain drop by 6 dB, giving a net drop in noise output of approximately 3 dB. This of course refers only to the internal noise of the amplifier stage, and pickup of external interference is always possible on an unterminated input. The input impedances here are modest, however, and the problem is less serious than you might think. Having said that, deliberately leaving inputs unterminated is always bad practice.

If this circuit is built with four 10 $\text{k}\Omega$ resistors and a 5532 op-amp section, the noise output is -104.8 dBu with the inputs terminated to ground via 50 Ω resistors. As noted above, the input impedance of the cold input is actually lower than the resistor connected to it when working balanced, and if it is desirable to raise this input impedance to 10 $\text{k}\Omega$, it could be done by raising the four resistors to 16 $\text{k}\Omega$; this slightly degrades the noise output to -103.5 dBu. Table 14.8 gives some examples of how the noise output depends on the resistor value; the third column gives the noise with the input unterminated, and shows that in each case the amplifier is about 3 dB quieter when open-circuited. It also shows that a useful improvement in noise performance is obtained by dropping the resistor values to the lowest that a 5532 can

TABLE 14.8 Noise output (dBu) measured from simple balanced amps using a 5532 section

R value (Ω)	50 Ω terminated inputs	Open-circuit inputs	Terminated/open difference
100k	-95.3	-97.8	2.5
10k	-104.8	-107.6	2.8
2k0	-109.2	-112.0	2.8
820	-111.7	-114.5	2.8

easily drive (the op-amp has to drive the feedback resistor), though this usually gives unacceptably low input impedances. More on that at the end of the chapter.

Variations on the Balanced Input Stage

I now give a collection of balanced input circuits that offer advantages or extra features over the standard balanced input configuration. The circuit diagrams often omit stabilizing capacitors, input filters, and DC-blocking capacitors to improve the clarity of the basic principle. They can easily be added; in particular, bear in mind that a stabilizing capacitor like C1 in Figure 14.7 is often needed between the op-amp output and the negative input to guarantee freedom from high-frequency oscillation.

Combined Unbalanced and Balanced Inputs

If both unbalanced and balanced inputs are required, it is extremely convenient if it can be arranged so that no switching between them is required. Switches cost money, mean more holes in the metalwork, and add to assembly time. Figure 14.8 shows an effective way to implement this. In balanced mode, the source is connected to the balanced input and the unbalanced input left unterminated. In unbalanced mode, the source is connected to the unbalanced input and the balanced input left unterminated, and no switching is required. It might appear that these unterminated inputs would pick up extra noise, but in practice this is not the case. It works very well and I have used it successfully in high-end equipment for two prestigious manufacturers.

As described above, in the world of hi-fi, balanced signals are at twice the level of the equivalent unbalanced signals, and so the balanced input must have a gain of 0.5 or -6 dB

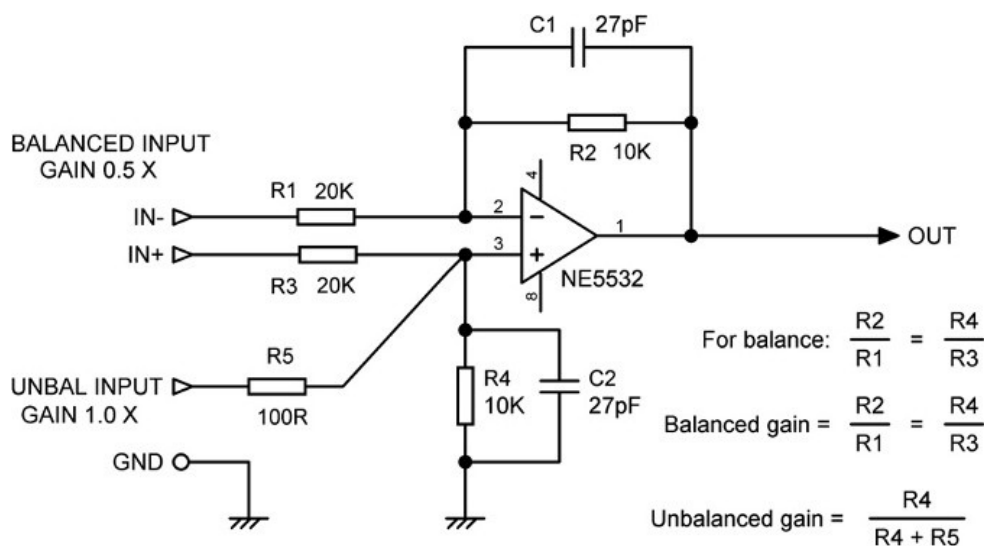


Figure 14.8: Combined balanced and unbalanced input amplifier with no switching required

relative to the unbalanced input to get the same gain by either path. This is done here by increasing R1 and R3 to 20 k Ω . The balanced gain can be greater or less than unity, but the gain via the unbalanced input is always 1. The differential gain of the amplifier and the constraints on the component values for balanced operation are shown in Figure 14.4, and are not repeated in the text to save space. This applies to the rest of the balanced inputs in this chapter.

There are two minor compromises in this circuit that need to be noted. Firstly, the noise performance in unbalanced mode is worse than for the dedicated unbalanced input described earlier in this chapter, because R2 is effectively in the signal path and adds Johnson noise. Secondly, the input impedance of the unbalanced input cannot be very high because it is set by R4, and if this is increased in value all the resistances must be increased proportionally and the noise performance will be markedly worse. It is important that only one input cable should be connected at a time, because if an unterminated cable is left connected to an unused input, the cable capacitance to ground can cause frequency response anomalies and might in adverse circumstances cause HF oscillation. A prominent warning on the back panel and in the manual is a very good idea.

The Superbal Input

This version of the balanced input amplifier, shown in Figure 14.9, has been referred to as the ‘Superbal’ circuit because it gives equal impedances into the two inputs for differential signals. It was originated by David Birt of the BBC [6]. With the circuit values shown the

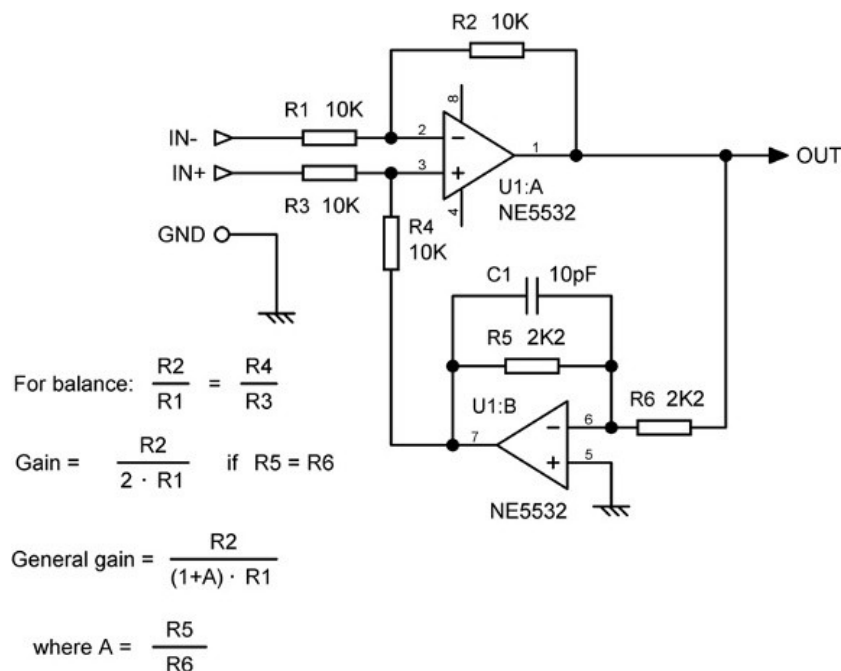


Figure 14.9: The Superbal balanced input amplifier

differential input impedance is exactly $10\text{ k}\Omega$ via both hot and cold inputs. The common-mode input impedance is $20\text{ k}\Omega$ as before.

In the standard balanced input R4 is connected to ground, but here its lower end is actively driven with an inverted version of the output signal, giving symmetry. The increased amount of negative feedback reduces the gain with four equal resistors to -6 dB instead of unity. The gain can be reduced below -6 dB by giving the inverter a gain of more than 1; if R1, R2, R3, and R4 are all equal, the gain is $1/(A + 1)$, where A is the gain of the inverter stage. This is of limited use as the inverter U1:B will now clip before the forward amplifier U1:A, reducing headroom. If the gain of the inverter stage is gradually reduced from unity to zero, the stage slowly turns back into a standard balanced amplifier with the gain increasing from -6 dB to unity and the input impedances becoming less and less equal. If a gain of less than unity is required it should be obtained by increasing R1 and R3.

R5 and R6 should be kept as low in value as possible to minimize Johnson noise; there is no reason why they have to be equal in value to R1, etc. The only restriction is the ability of U1:A to drive R6 and U1:B to drive R5, both resistors being effectively grounded at one end. The capacitor C1 will almost certainly be needed to ensure HF stability; the value in the figure is only a suggestion. It should be kept as small as possible because reducing the bandwidth of the inverter stage impairs CMRR at high frequencies.

Switched-Gain Balanced Inputs

A balanced input stage that can be switched to two different gains while maintaining CMRR is very useful. Equipment often has to give optimal performance with both semi-pro (-7.8 dBu) and professional ($+4\text{ dBu}$) input levels. If the nominal internal level of the system is in the normal range of -2 to -6 dBu , the input stage must be able to switch between amplifying and attenuating, while maintaining good CMRR in both modes.

The brute-force way to change gain in a balanced input stage is to switch the values of either R1 and R3, or R2 and R4, in Figure 14.4, keeping the pairs equal in value to maintain the CMRR; this needs a double-pole switch for each input channel. A much more elegant technique is shown in Figure 14.11. Perhaps surprisingly, the gain of a differential amplifier can be manipulated by changing the drive to the feedback arm (R2 etc.) only, and leaving the other arm R4 unchanged, without affecting the CMRR. The essential point is to keep the source resistance of the feedback arm the same, but drive it from a scaled version of the op-amp output. Figure 14.10 does this with the network R5, R6, which has a source resistance made up of $6\text{ k}\Omega$ in parallel with $2\text{ k}\Omega$, which is $1.662\text{ k}\Omega$. This is true whether R6 is switched to the op-amp output (low gain setting) or to ground (high gain setting), for both have effectively zero impedance. For low gain the negative feedback is not attenuated, but fed through to R2 and R7 via R5, R6 in parallel. For high gain R5 and R6 become a potential divider, so the

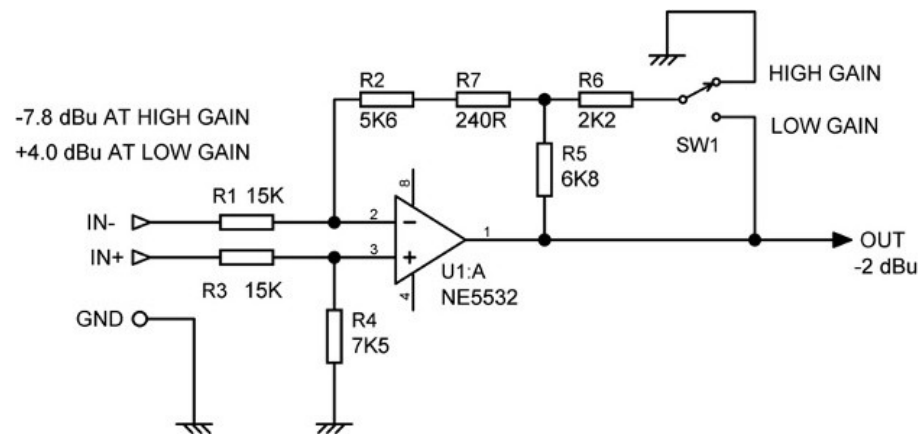


Figure 14.10: A balanced input amplifier with gain switching that maintains good CMRR

amount of feedback is decreased and the gain increased. The value of $R2 + R7$ is reduced from $7k5$ by $1.662 k\Omega$ to allow for the source impedance of the $R5, R6$ network; this requires the distinctly non-standard value of $5.838 k\Omega$ that is here approximated by $R2$ and $R7$, which give $5.6 k\Omega + 240 \Omega = 5.840 k\Omega$. This value is the best that can be done with E24 resistors; it is obviously out by 2Ω , but that is much less than a 1% tolerance on $R2$, and so will have only a vanishingly small effect on the CMRR.

Note that this stage can attenuate as well as amplify if $R1, R3$ are set to be greater than $R2, R4$, as shown here. The nominal output level of the stage is assumed to be -2 dBu; with the values shown the two gains are -6.0 and $+6.2$ dB, so $+4$ and -7.8 dBu respectively will give -2 dBu at the output. Other pairs of gains can of course be obtained by changing the resistor values; the important thing is to stick to the principle that the value of $R2 + R7$ is reduced from the value of $R4$ by the source impedance of the $R5, R6$ network. With the values shown the differential input impedance is $11.25 k\Omega$ via the cold and $22.5 k\Omega$ via the hot input. The common-mode input impedance is $22.5 k\Omega$.

This neat little circuit has the added advantage that nothing bad happens when the switch is moved with the circuit operating. When the wiper is between contacts you simply get a gain intermediate between the high and low settings, which is pretty much the ideal situation.

Variable-Gain Balanced Inputs

The beauty of a variable-gain balanced input is that it allows you to get the incoming signal up or down to the nominal internal level as soon as possible, minimizing both the risk of clipping and contamination with circuit noise. The obvious method of making a variable-gain differential stage is to use dual-gang pots to vary either $R1, R3$ or $R2, R4$ together, to maintain CMRR. This is clumsy, and gives a CMRR that is both bad and highly variable due to the

inevitable mismatches between pot sections. For a stereo input the required four-gang pot is an unappealing proposition.

There is, however, a way to get a variable gain with good CMRR, using a single pot section. The principle is essentially the same as for the switched-gain amplifier above: keep constant the source impedance driving the feedback arm, but vary the voltage applied. The principle is shown in Figure 14.11. To the best of my knowledge I invented this circuit in 1982; any comments on this point are welcome. The feedback arm R2 is driven by voltage-follower U1:B. This eliminates the variations in source impedance at the pot wiper, which would badly degrade the CMRR. R6 limits the gain range and R5 modifies the gain law to give it a more usable shape. When the pot is fully up (minimum gain) R5 is directly across the output of U1:A so do not make it too low in value. If a center-detent pot is used to give a default gain setting, this may not be very accurate as it partly depends on the ratio of pot track (no better than $\pm 10\%$ tolerance, and sometimes worse) to 1% fixed resistors.

This configuration is very useful as a general line input with an input sensitivity range of -20 to $+10$ dBu. For a nominal output of 0 dBu, the gain of Figure 14.11 is $+20$ to -10 dB, with R5 chosen for 0 dB gain at the central wiper position. An op-amp in a feedback path may appear a dubious proposition for HF stability, because of the extra phase-shift it introduces, but here it is working as a voltage-follower, so its bandwidth is maximized and in practice the circuit is dependably stable.

High-Input-Impedance Balanced Inputs

We saw earlier that high input impedances are required to maximize CMRR of a balanced interconnection, but the input impedances offered by the standard balanced circuit are limited by the need to keep the resistor values down to control Johnson noise. High-impedance balanced inputs are also useful for interfacing to valve equipment in the strange world of retro

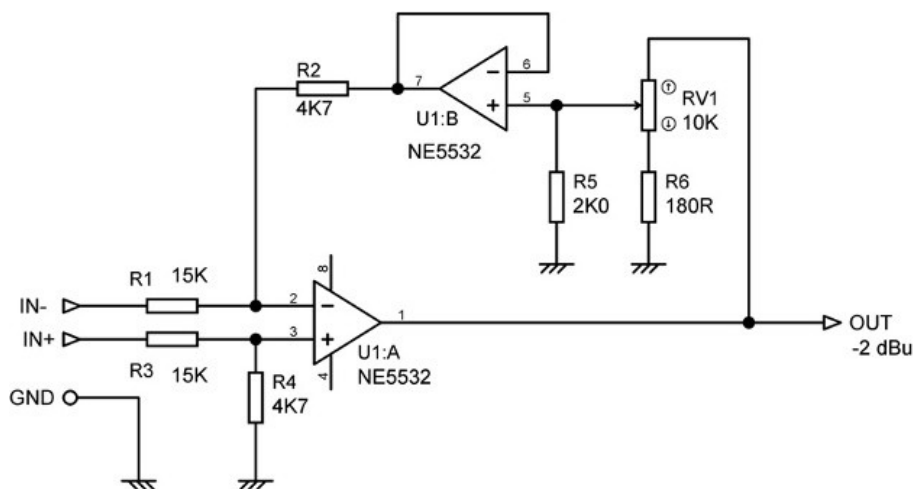


Figure 14.11: Variable-gain balanced input amplifier

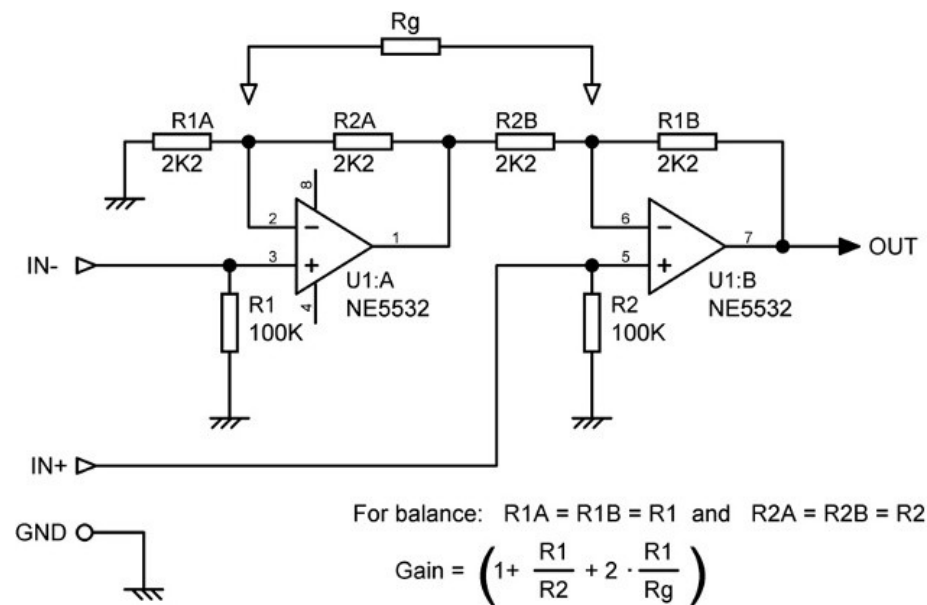


Figure 14.12: High-input-impedance balanced input

hi-fi. Adding output cathode-followers to valve circuitry is expensive and consumes a lot of extra power, and so the output is often taken directly from the anode of a gain stage, and even a so-called bridging load of 10 k Ω may seriously compromise the distortion performance and output capability of the source equipment.

Figure 14.12 shows a configuration where the input impedances are determined only by the bias resistances R1 and R2. They are shown here as 100 k Ω , but may be considerably higher if op-amp bias currents permit. A useful property of this circuit is that adding a single resistor R_g increases the gain, but preserves the circuit balance and CMRR. This configuration cannot be set to attenuate because the gain of an op-amp with series feedback cannot be reduced below unity.

It is of course always possible to give a basic balanced input a high input impedance by putting unity-gain buffers in front of each input, but that uses three op-amp sections rather than two. Sometimes, however, it is appropriate. Much more on that later.

We noted earlier that the simple balanced input is surprisingly quiet and well behaved when its inputs are unterminated. This is not the case with this configuration, which because of its high input impedances will be both noisy and susceptible to picking up external interference.

The Inverting Two-Op-Amp Input

The configuration depicted in Figure 14.13 has its uses because the hot and cold inputs have the same impedances for differential signals, as well as for common-mode voltages [7]. It is not suited to high input impedances at normal gains because high resistor values would have to be

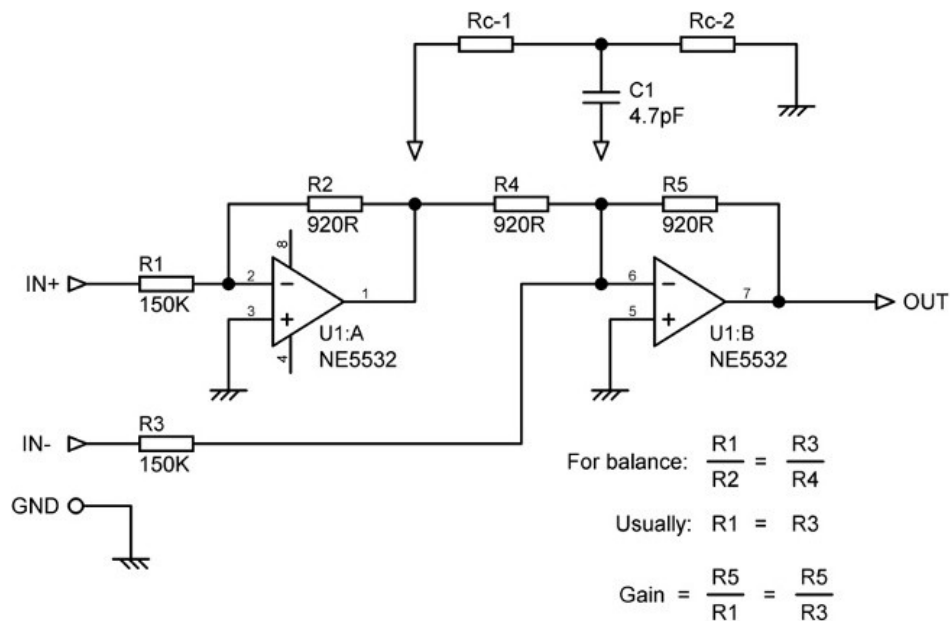


Figure 14.13: Inverting two-op-amp balanced input. Values shown interface a 100 V line to -2 dBu internal level

used throughout and they would generate excess Johnson noise, but if it is interfacing with a high voltage source so the gain must be well below unity, $R1$ and $R3$ can be made high in value, and $R2$, $R4$, $R5$ set low, the latter components keeping the noise down. The CMRR may degrade at HF because the hot signal has gone through an extra op-amp and suffered phase-shift, interfering with the subtraction; this can be compensated for by the network $Rc-1$, $Rc-2$, and $C1$; the values needed depend on op-amp type and must be checked by CMRR measurements.

I have used this configuration for balanced input feeds from a 100 V loudspeaker line of the sort still in use for distributing audio over wide areas. The circuit values shown reduce the 100 V input to a nominal -2 dBu (615 mV) internal level. If this circuit is used for that purpose then effective input overvoltage protection on the inputs is essential as large voltage transients are possible on a loudspeaker line if parts of it are unplugged or plugged with signal present, due to the inductance of the line-matching transformers. This protection can be conveniently provided by the usual diode clamps to the supply rails; this is dealt with at the end of this chapter. The input resistors will be of high value in this application so there is very little possibility of excessive inputs overheating the resistors or ‘pumping up’ the op-amp supply rails.

The Instrumentation Amplifier

Almost every book on balanced or differential inputs includes the three-op-amp circuit of Figure 14.14 and praises it as the highest expression of the differential amplifier. It is usually called the instrumentation amplifier configuration because of its undoubted superiority for data acquisition. (Specialized ICs exist that are sometimes also called instrumentation

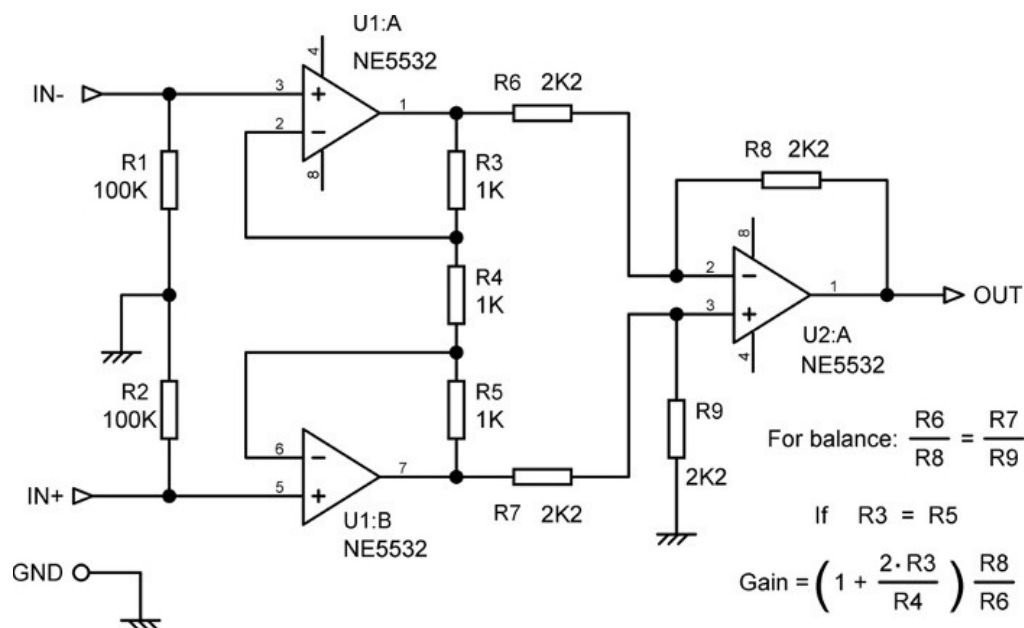


Figure 14.14: The instrumentation amplifier configuration. Gain here is 3 times

amplifiers or in-amps; these are designed for very high CMRR data acquisition. They are expensive and in general not optimized for audio work.)

The beauty of the three-op-amp configuration is that the dual input stage buffers the balanced line from the input impedances of the final differential amplifier; the four resistances around it can now be made much lower in value, reducing their Johnson noise by a significant amount, while keeping the CMRR benefits of presenting high input impedances to the balanced line. The other feature, which is usually much more emphasized because of its unquestionable elegance, is that the dual input stage, with its shared feedback network $R3$, $R4$, $R5$, can be set to have a high differential gain by giving $R4$ a low value, but its common-mode gain is always unity; this property is not affected by mismatches in $R3$ and $R5$. The final amplifier then does its usual job of common-mode rejection, and the combined CMRR can be very good indeed if the first stage gain is high.

This is all well and good, but as it stands this stage is not very useful for audio balanced line inputs. A data-acquisition application like ECG monitoring may need a gain of thousands of times, which will allow a stunning CMRR to be achieved without using precision resistors, but the cruel fact is that, in audio usage, gain at this point is very often simply not wanted. In a typical signal path comprised of op-amps, the nominal internal level is usually between -6 and 0 dBu, and the level coming in is at the professional level of $+4$ dBu; what is needed is 6 dB of attenuation rather than any gain. Gain at this point and attenuation later would introduce what can only be described as a headroom bottleneck. If the incoming level was the semi-pro -7.8 dBu a small amount of gain could be introduced, but then the CMRR advantage would be equally small, and certainly not worth the cost of two op-amp sections.

Transformer Balanced Inputs

When it is essential that there is no galvanic connection (i.e. no electrical conductor) between two pieces of equipment, transformer inputs are indispensable. They are also useful if EMC conditions are severe. Figure 14.15 shows a typical transformer input. The transformer usually has a 1:1 ratio, and is enclosed in a metal shielding can that must be grounded. Good line transformers have an interwinding shield that must also be grounded or the high-frequency CMRR will be severely compromised. The transformer secondary must see a high impedance as this is reflected to the primary and represents the input impedance; here it is set by R2, and a buffer drives the circuitry downstream. In addition, if the secondary loading is too heavy there will be increased transformer distortion at low frequencies. Line-input transformers are built with small cores and are only intended to deliver very small amounts of power; they are *not* interchangeable with line-output transformers. A most ingenious approach to dealing with this distortion problem by operating the input transformer core at near-zero flux was published by Paul Zwicky in 1986 [8].

There is a bit more to correctly loading the transformer secondary. If it is simply loaded with a high-value resistor there will be peaking of the frequency response due to resonance between the transformer leakage inductance and the winding capacitance. This is shown in Figure 14.16, where a Sowter 3276 line input transformer (a high-quality component) was given a basic resistive loading of 100 k Ω . The result was trace A, which has a 10 dB peak around 60 kHz. This is bad not only because it accentuates the effect of out-of-band noise, but because it intrudes on the audio frequency response, giving a lift of 1 dB at 20 kHz. Reducing the resistive load R2 would damp the resonance, but it would also reduce the input impedance. The answer is to add a Zobel network, i.e. a resistor and capacitor in series, across the secondary; this has no effect except at high frequencies. The first attempt used R1 = 2k7 and C1 = 1 nF, giving trace B, where the peaking has been reduced to 4 dB around 40 kHz, but the 20 kHz lift is actually slightly greater. R1 = 2k7 and C1 = 2 nF gave trace C, which is a bit

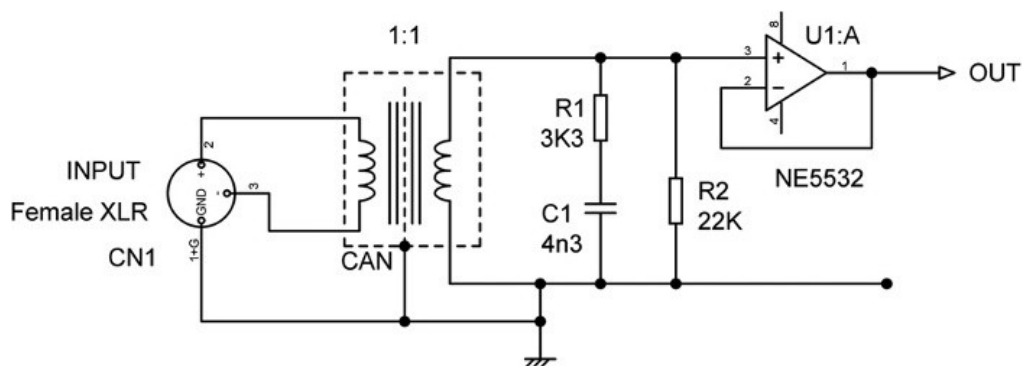


Figure 14.15: A transformer balanced input. R1 and C1 are the Zobel network that damps the secondary resonance

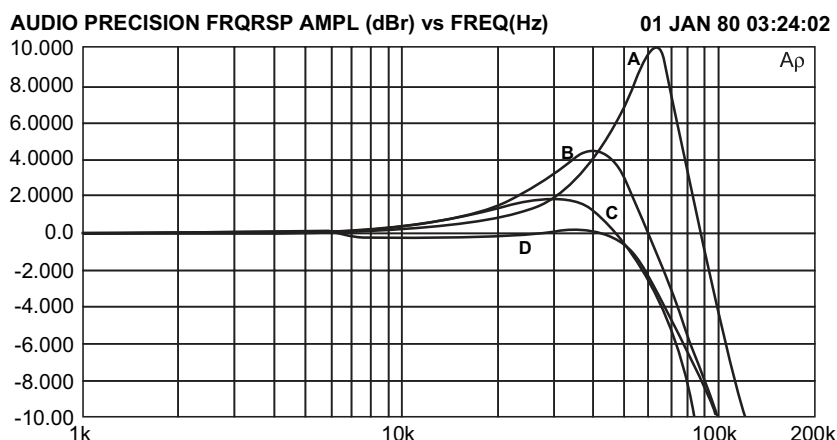


Figure 14.16: Optimizing the frequency response of a transformer balanced input with a Zobel network

better in that it only has a 2 dB peak. A bit more experimentation ended up with $R1 = 3k\Omega$ and $C1 = 4.3 \text{ nF}$ ($3n3 + 1 \text{ nF}$) and yielded trace D, which is pretty flat, though there is a small droop around 10 kHz. The Zobel values are fairly critical for the flattest possible response, and must certainly be adjusted if the transformer type is changed.

Input Overvoltage Protection

Input overvoltage protection is not common in hi-fi applications, but is regarded as essential in most professional equipment. The normal method is to use clamping diodes, as shown in [Figure 14.18](#), that prevent certain points in the input circuitry from moving outside the supply rails.

This is straightforward, but there are two points to watch. Firstly, the ability of this circuit to withstand excessive input levels is not without limit. Sustained overvoltages may burn out $R5$ and $R6$, or pump unwanted amounts of current into the supply rails; this sort of protection is mainly aimed at transients. Secondly, diodes have a non-linear junction capacitance when they are reverse biased, so if the source impedance is significant the diodes will cause distortion at high frequencies. To quantify this problem here are the results of a few tests. If [Figure 14.17](#) is fed from the low impedance of the usual kind of line-output stage, the impedance at the diodes will be about $1 \text{ k}\Omega$ and the distortion induced into an $11 \text{ V}_{\text{rms}}$, 20 kHz input will be below the noise floor. However, if the source impedance is high so the impedance at the diodes is increased to $10 \text{ k}\Omega$, with the same input level, the THD at 20 kHz is degraded from 0.0030% to 0.0044% by adding the diodes. I have thought up a rather elegant way to eliminate this effect completely, but this is not the place to disclose it.

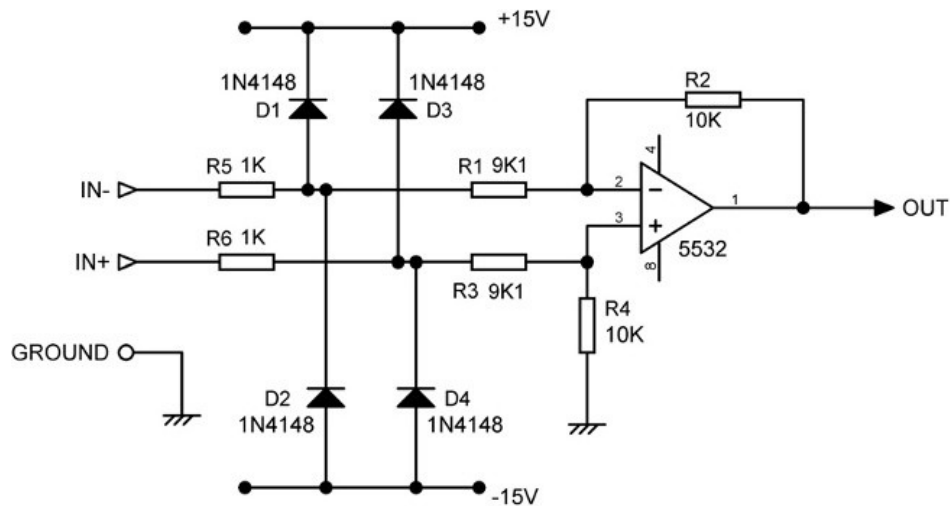


Figure 14.17: Input overvoltage protection for a balanced input amplifier

Low-Noise Balanced Inputs

So far we have not said much about the noise performance of balanced inputs, though on our way through the chapter we have remarked several times that the standard balanced input amplifier with four 10 k Ω resistors shown in Figure 14.18(a) is markedly noisier than an unbalanced input like that in Figure 14.1. The unbalanced input stage, with its input terminated by 50 Ω to ground, has a noise output of -119.0 dBu over the usual 22 Hz – 22 kHz bandwidth. If the balanced circuit is built with 10 k Ω resistors and a 5532 section, the noise output is -104.8 dBu with the inputs similarly terminated. This is a big difference of 14.2 dB.

In the hi-fi world in particular, where an amplifier may have both unbalanced and balanced inputs, many people would feel that this situation is the wrong way round. Surely the balanced input, with its professional XLR connector and its much-vaunted rejection of ground noise, should show a better performance in all departments? Well, it does – except as regards internal noise, and a 14 dB discrepancy is both clearly audible and hard to explain away. This section explains how to design it away instead.

We know that the source of the extra noise is the relatively high resistor values around the op-amp (see Table 14.8 earlier in the chapter), but these cannot be reduced in the simple balanced input without reducing the input impedances below what is acceptable. The answer is to lower the resistor values but buffer them from the input with a pair of voltage-followers. This arrangement is shown in Figure 14.18(b); 5532s are a good choice for this as they combine low voltage noise with low distortion and good load-driving capability. Since the input buffers are voltage-followers with 100% feedback, their gain is very accurately defined at unity and CMRR is not degraded; CMRR is still defined by the resistor tolerances, and by the bandwidth of the differential op-amp.

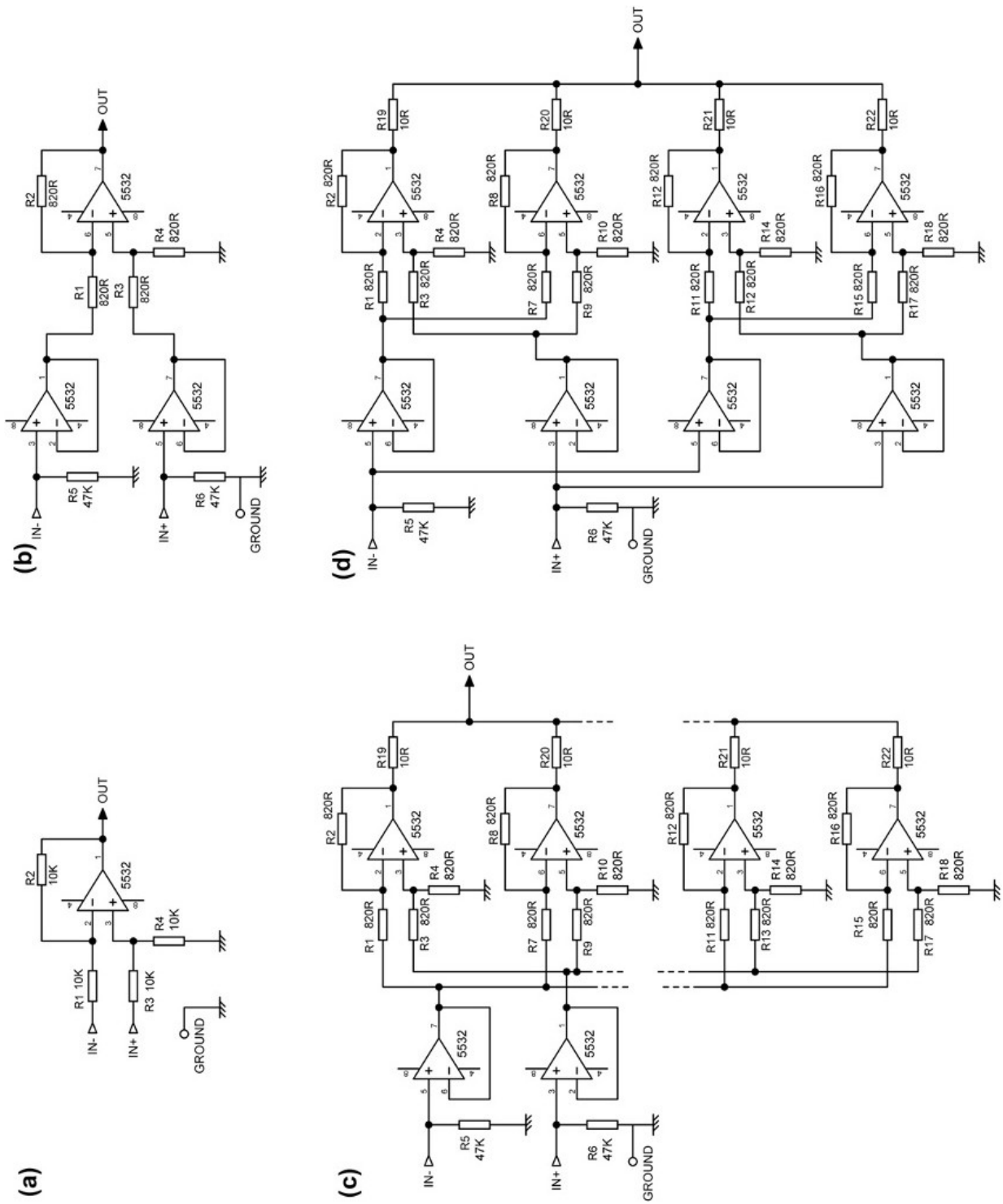


Figure 14.18: Low-noise unity-gain balanced inputs using multiple 5532 buffers and differential amplifiers

There is a limit to how far the four resistors can be reduced, as the differential stage has to be driven by the input buffers, and it also has to drive its own feedback arm. If 5532s are used a safe value that gives no measurable deterioration of the distortion performance is about $820\ \Omega$, and a 5532 differential stage alone (without the buffers) and $4 \times 820\ \Omega$ resistors gives a noise output of $-111.7\ \text{dBu}$, which is 6.6 dB lower than the standard $4 \times 10\ \text{k}\Omega$ version. Adding the two input buffers degrades this only slightly to $-110.2\ \text{dB}$, because we are adding only the voltage noise component of the two new op-amps, and we are still 5.1 dB quieter than the original $4 \times 10\ \text{k}\Omega$ version. It is interesting that we now have three op-amps in the signal path instead of one, but we still get a significantly lower noise level.

This might appear to be all we can do; it is not possible to reduce the value of the four resistors around the differential amplifier any further without compromising linearity. However, there is almost always some way to go further in the great game that is electronics, and here are three possibilities. A step-up transformer could be used to exploit the low source impedance (remember we are still assuming the source impedances are $50\ \Omega$) and it might well work superbly in terms of noise alone, but transformers are always heavy, expensive, susceptible to magnetic fields, and of doubtful low-frequency linearity. We would also very quickly run into headroom problems; balanced line input amplifiers are normally required to attenuate rather than amplify.

We could design a discrete op-amp hybrid stage with discrete input transistors, which are quieter than those integrated into IC op-amps, coupled to an op-amp to provide raw loop gain; this can be quite effective but you need to be very careful about high-frequency stability, and it is difficult to get an improvement of more than 6 dB. Thirdly, we could design our own op-amp using all discrete parts; this approach tends to have fewer stability problems as all circuit parameters are accessible, but it definitely requires rather specialized skills, and the result takes up a lot of PCB area.

Since none of those three approaches is appealing, now what? One of the most useful techniques in low-noise electronics is to use two identical amplifiers so that the gains add arithmetically, but the noise from the two separate amplifiers, being uncorrelated, partially cancels. Thus, we get a 3 dB noise advantage each time the number of amplifiers used is doubled. This technique works very well with multiple op-amps, as expounded in Chapter 1; let us apply it and see how far it may be usefully taken.

Since the noise of a single 5532-section unity-gain buffer is only $-119.0\ \text{dBu}$, and the noise from the $4 \times 820\ \Omega$ differential stage (without buffers) is a much higher $-111.7\ \text{dBu}$, the differential stage is clearly the place to start work. We will begin by using two identical $4 \times 820\ \Omega$ differential amplifiers as shown in the top section of [Figure 14.18\(c\)](#), both driven from the existing pair of input buffers. This will give partial cancellation of both resistor and op-amp noise from the two differential stages if their outputs are summed. The main question is how to

sum the two amplifier outputs; any active solution would introduce another op-amp, and hence more noise, and we would almost certainly wind up worse off than when we started. The answer is, however, beautifully simple. We just connect the two amplifier outputs together with $10\ \Omega$ resistors; the gain does not change but the noise output drops. The signal output of both amplifiers is nominally the same, so no current should flow from one op-amp output to the other. In practice there will be slight gain differences due to resistor tolerances, but with 1% resistors I have never experienced any hint of a problem. The combining resistor values are so low at $10\ \Omega$ that their Johnson noise contribution is negligible.

The use of multiple differential amplifiers has another advantage – the CMRR errors are also reduced in the same way that the noise is reduced. This is also similar to the use of multiple parallel capacitors to improve RIAA accuracy, as explained in Chapter 7.

We therefore have the arrangement of [Figure 14.18\(c\)](#), with single input buffers (i.e. one per input) and two differential amplifiers, and this reduces the noise output by 2.3 dB to -112.5 dBu, which is quieter than the original $4 \times 10\ \text{k}\Omega$ version by an encouraging 7.4 dB. We do not get a full 3 dB noise improvement because both differential amplifiers are handling the noise from the input buffers, which is correlated and so is not reduced by partial cancellation. The contribution of the input buffer noise is further brought out if we take the next step of using four differential amplifiers. There is of course nothing special about using amplifiers in powers of two. It is perfectly possible to use three or five differential amplifiers in the array, which will give intermediate amounts of noise reduction. If you have a spare op-amp section, then put it to work!

So, leaving the input buffers unchanged, we use them to drive an array of four differential amplifiers. These are added on at the dotted lines in the lower half of [Figure 14.18\(c\)](#). We get a further improvement, but only by 1.5 dB this time. The output noise is down to -114.0 dBu, quieter than the original $4 \times 10\ \text{k}\Omega$ version by 8.9 dB. You can see that at this point we are proceeding by decreasing steps, as the input buffer noise is starting to dominate, and there seems little point in doubling up the differential amplifiers again; the amount of hardware could be regarded as a bit excessive, and so would the PCB area occupied. The increased loading on the input buffers is also a bit of a worry.

A more fruitful approach is to tackle the noise from the input buffers, by doubling them up as in [Figure 14.18\(d\)](#), so that each buffer drives only two of the four differential amplifiers. This means that the buffer noise will also undergo partial cancellation, and will be reduced by 3 dB. There is, however, still the contribution from the differential amplifier noise, and so the actual improvement on the previous version is 2.2 dB, bringing the output noise down to -116.2 dBu, which is quieter than the original $4 \times 10\ \text{k}\Omega$ version by a thumping 11.1 dB. Remember that there are two inputs, and ‘double buffers’ means two buffers per hot and cold input, giving a total of four in the complete circuit.

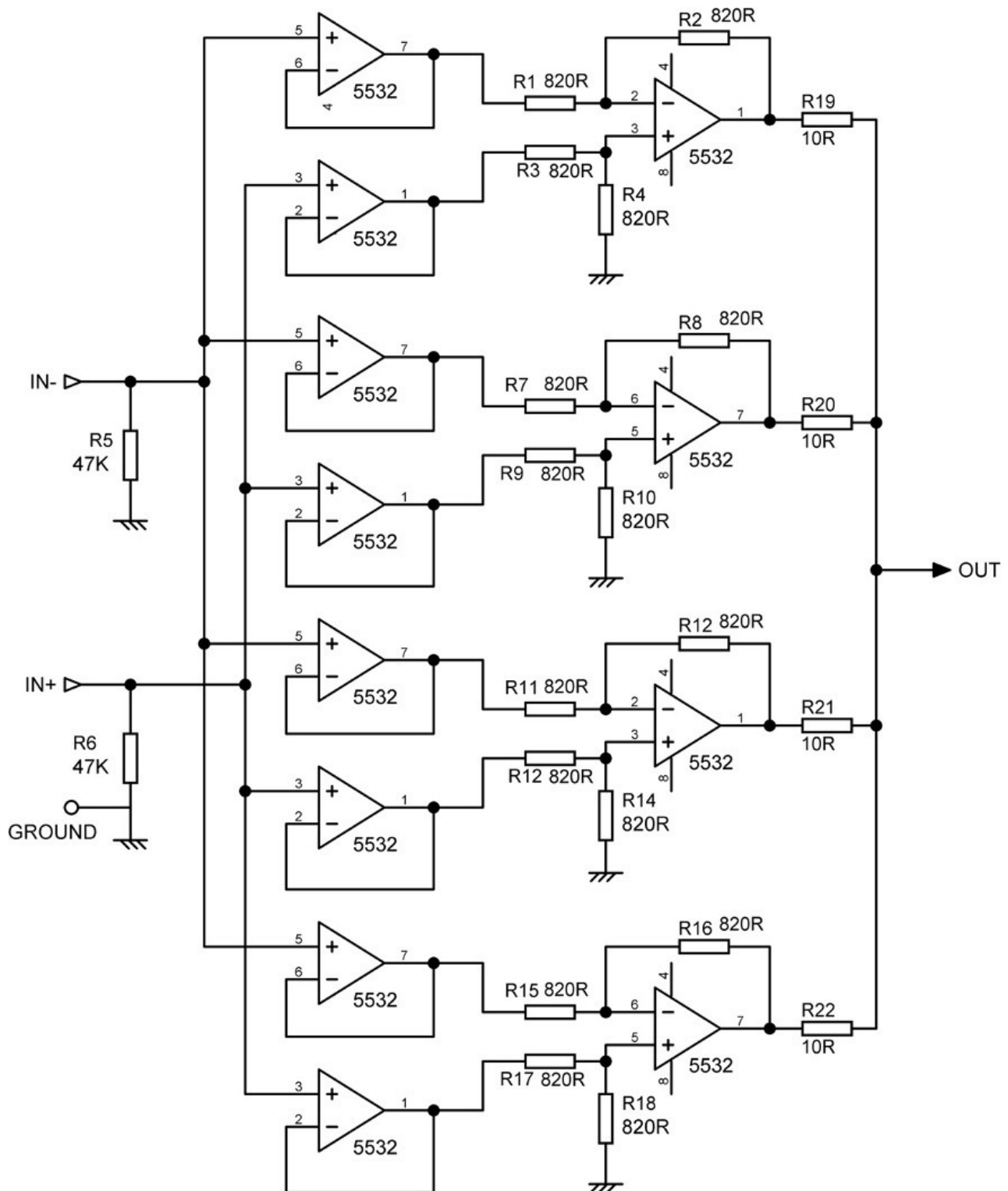


Figure 14.19: The final 5532 low-noise unity-gain balanced input stage, with quad input buffers and four differential amplifiers. The noise output is only -117.0 dBu

Since doubling up the input buffers gave us a useful improvement, it's worth trying again, so we have a structure with quad buffers and four differential amplifiers, as shown in [Figure 14.19](#), where each differential amplifier now has its very own buffer. This improves on the previous version by a rather less satisfying 0.8 dB, giving an output noise level of -117.0 dBu, quieter than the original 4×10 k Ω version by 11.9 dB. The small improvement we have gained indicates that the focus of noise reduction needs to be returned to the differential amplifier array, but the next step would seem to be using eight amplifiers, which is not very appealing. Thoughts about ears of corn on chessboards tend to intrude at this point.

This is a good moment to pause and see what we have achieved. We have built a balanced input stage that is quieter than the standard circuit by 11.9 dB, using standard components of low cost. We have used increasing numbers of them, but the total cost is still small compared with enclosures, power supplies, etc. On the other hand, the power consumption is clearly several times greater. The technology is highly predictable and the noise reduction reliable, in fact bulletproof. The linearity is as good as that of a single op-amp of the same type, and in the same way there are no HF stability problems.

What we have not done is build a balanced input that is quieter than the unbalanced one – we are still 2.0 dB short of that target, but at least we have reached a point where the balanced input is not obviously noisier.

These noise results are summarized in [Table 14.10](#) at the end of the chapter.

Low-Noise Balanced Inputs in Real Life

Please don't think that this examination of low-noise input options is merely a voyage off into pure theory. It has real uses and has been applied in practice. The Cambridge Audio 840W power amplifier is a design of mine which, I might modestly mention in passing, won a CES Innovation Award in January 2008. This unit has both unbalanced and balanced inputs, and conventional technology would have meant that the balanced inputs would have been significantly the noisier of the two. Since the balanced input is the 'premium' input, many people would think there was something amiss with this state of affairs. We therefore decided the balanced input had to be quieter than the unbalanced input. Using 5532s in an architecture similar to those outlined above, this requirement proved straightforwardly attainable, and the final balanced input design was both economical and quieter than its unbalanced neighbor by a dependable 0.9 dB. Two other versions were evaluated that made the balanced input quieter than the unbalanced one by 2.8 dB, and by 4.7 dB, at somewhat greater cost and complexity. These were put away for possible future upgrades.

The Signal Transfer Company [\[9\]](#) manufactures a low-noise balanced input card based on these principles that has 47 k Ω input impedances, unity gain, and a noise output of only -115 dBu.

Ultra-Low-Noise Balanced Inputs

In the section on low-noise balanced inputs above, we reclined briefly on our laurels, having achieved an economical balanced input stage with output noise at the extremely low level of -117.0 dBu. Regrettably this is still 2 dB noisier than a simple unbalanced input. It would be wrong to conclude from this that the resources of electronic design are exhausted. At the end of the noise-reduction sequence we were aware that the dominant noise source was currently the differential amplifier array, and we shrank from doubling up again to use eight amplifiers because of issues of cost and the PCB area occupied. We will take things another step by taking a much more relaxed view of cost (as one does at the ‘high end’) and see how that changes the game. We will, however, retain some concern about PCB area.

An alternative way to make the differential amplifier array quieter is simply to use op-amps that are quieter. These will inevitably be more expensive – much more expensive – than the ubiquitous 5532. Because of the low resistor values around the op-amps, we need to focus on low voltage noise rather than low current noise, and there are several that are significantly better than the 5532, as shown by the typical noise density figures in Table 14.9.

Clearly, moving to the 5534A will give a significant noise reduction, but since there is only a single op-amp per package, and external compensation is needed, the board area used will be much greater. The new chip on the block is the LM4562, a bipolar op-amp that has finally surpassed the 5532 in performance. The input voltage noise density is typically 2.7 nV/ $\sqrt{\text{Hz}}$, substantially lower than the 5 nV/ $\sqrt{\text{Hz}}$ of the 5532. For applications with low source impedances, this implies a handy noise advantage of 5 dB or more. The LM4562 is a dual op-amp and will not take up more space. At the time of writing it is something like 10 times more expensive than the 5532.

Step 1 – we replace all four op-amps in the differential amplifiers with LM4562s. They are a drop-in replacement with no circuit adjustments required at all. We leave the quad 5532 input buffers in place. The noise output drops by an impressive 1.9 dB, giving an output noise level of -118.9 dBu, quieter than the original 4×10 k Ω version by 14.1 dB, and only 0.1 dB noisier than the unbalanced stage.

TABLE 14.9 Voltage and noise densities for low-noise balanced-input op-amp candidates

Op-amp	Voltage noise density (nV/ $\sqrt{\text{Hz}}$)	Current noise density (pA/ $\sqrt{\text{Hz}}$)
5532	5	0.7
5534A	3.5	0.4
LM4562	2.7	1.6
AD797	0.9	2
LT1028	0.85	1

Step 2 – replace the quad 5532 buffers with quad LM4562 buffers. Noise falls by only 0.6 dB, the output being -119.5 dBu, but at last we have a balanced stage that is quieter than the unbalanced stage, by a small but solid 0.5 dB.

One of the pre-eminent low-noise-with-low-source-resistance op-amps is the AD797 from Analog Devices, which has a remarkably low voltage noise of $0.9 \text{ nV}/\sqrt{\text{Hz}}$ (typical at 1 kHz) but it is a very expensive part, costing between 20 and 25 times more than a 5532 at the time of writing. The AD797 is a single op-amp, while the 5532 is a dual, so the cost per op-amp is actually 40–50 times greater, and more PCB area is required, but the potential improvement is so great we will overlook that.

Step 3 – we replace all four op-amps in the differential amplifiers with AD797s, putting the 5532s back into the input buffers in the hope that we might be able to save money somewhere. The noise output drops by a rather disappointing 0.4 dB, giving an output noise level of -119.9 dBu, quieter than the original $4 \times 10 \text{ k}\Omega$ version by 15.1 dB.

Perhaps putting those 5532s back in the buffers was a mistake? Our fourth and final move in this game of electronic chess is to replace all the quad 5532 input buffers with dual (not quad) AD797 buffers. This requires another four AD797s (two per input) and is once more not a cheap strategy. We retain the four AD797s in the differential amplifiers. The noise drops by another 0.7 dB, yielding an output noise level of -120.6 dBu, quieter than the original $4 \times 10 \text{ k}\Omega$ version by 15.8 dB, and quieter than the unbalanced stage by a satisfying 1.6 dB. You can do pretty much anything in electronics with a bit of thought and a bit of money.

You are probably wondering what happened to the LT1028 lurking at the bottom of [Table 14.9](#). It is true that its voltage noise density is slightly better than that of the AD797, but there is a subtle snag. As described in Chapter 7, the LT1028 has bias-current cancellation circuitry that injects correlated noise currents into the two inputs. These will cancel if the impedances seen by the two inputs are the same, but in moving-magnet amplifier use the impedances differ radically and the LT1028 is not useful in this application. The input conditions here are more benign, but the extra complication is unwelcome and I have never used the LT1028 in audio work. In addition, it is a single op-amp with no dual version.

This is not of course the end of the road. The small noise improvement in the last step we made tells us that the differential amplifier array is still the dominant noise source, and further development would have to focus on this. A first step would be to see if the relatively high current noise of the AD797s is significant with respect to the surrounding resistor values. If so, we need to see if the resistor values can be reduced without degrading linearity at full output. We should also check the Johnson noise contribution of all those $820 \text{ }\Omega$ resistors; they are generating -123.5 dBu each at room temperature, but of course the partial cancellation effect applies to them as well.

All these noise results are also summarized in [Table 14.10](#).

TABLE 14.10 A summary of the noise improvements made to the balanced input stage

Buffer type	Amplifier	Noise output (dBU)	Improvement on previous version (dB)	Improvement over $4 \times 10 \text{ k}\Omega$ diff. amp (dB)	Noisier than unbalanced input by: (dB)
	5532 voltage-follower	-119.0			0.0 dB ref.
None	Standard diff. amp 10K 5532	-104.8	0	0.0 dB ref.	14.2
None	Single diff. amp 820R 5532	-111.7	6.9	6.9	7.3
Single 5532	Single diff. amp 820R 5532	-110.2	5.4	5.4	8.8
Single 5532	Dual diff. amp 820R 5532	-112.50	2.3	7.4	6.5
Single 5532	Quad diff. amp 820R 5532	-114.00	1.5	8.9	5.0
Dual 5532	Quad diff. amp 820R 5532	-116.20	2.2	11.1	2.8
Quad 5532	Quad diff. amp 820R 5532	-117.00	0.8	11.9	2.0
Quad 5532	Quad diff. amp 820R LM4562	-118.9	1.9	14.1	0.1
Quad LM4562	Quad diff. amp 820R LM4562	-119.5	0.6	14.7	-0.5
Quad 5532	Quad diff. amp 820R AD797	-119.9	0.4	15.1	-0.9
Dual AD797	Quad diff. amp 820R AD797	-120.6	0.7	15.8	-1.6

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